

DECISION TREES: Directed Structure comprised of

nodes:
 → Each node → Test on attribute
 → Each branch → corresponds to attribute value/condition
 → leaves represent a class

⇒ Example of Inductive learning - specific example
se general rule

* Choosing splitting attribute? nikaralra

→ Info Gain → Gain Ratio → Gain measure

Info Gain measures how much information an attribute gives about the class -

→ Attr. that perfectly partition give maximal information

→ unrelated attributes should give no info.

→ It measures reduction in entropy

$$S = \text{Entropy} = -P_{\oplus} \log_2 P_{\oplus} - P_{\ominus} \log_2 P_{\ominus}$$

$P_{\oplus}$ = proportion of +ve examples

$P_{\ominus}$ = proportion of -ve examples

- High Entropy → Uniform distribution
- Low Entropy → Varied distribution

Minimum bits needed for c classes:

$$H(Y) = - \sum_{i=1}^c p_i \log_2 p_i$$

$$H(Y|X_K) = \sum_{j=1}^m P(X_K = v_j) H(Y|X_K = v_j)$$

$$\text{Info Gain} = H(Y) - H(Y|X_K)$$

Example:

	School	Football likers
1	1	Y
2	2	N
3	1	N
4	1	Y
5	1	N
6	3	Y
7	2	N
8	3	Y

Solution

v_j	$P(X = v_j)$	IG
1	$4/8 = 0.5$	1
2	$2/8 = 0.25$	0
3	$2/8 = 0.25$	0

log formula
↑

$$H(Y) = -\frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8} = 1$$

$$\text{IG}(Y|X_K) = H(Y) - H(Y|X_K) = 1 - 0.5 = 0.5$$

$\text{Info Gain} = \text{Entropy before split} - \text{Entropy after split}$

LAST MOMENT TUTORIALS LECTURE:

Age	comp ^{attributes}	Type ^{attributes}	Profit ^{→ class attribute}
old	Yes	Soft	Down
old	No	Soft	Down
old	No	Hard	Down
mild	Yes	Soft	Down
mild	Yes	Hard	Down
mild	No	Hard	Up
mild	No	Soft	Up
new	Yes	Soft	Up
new	No	Hard	Up
new	No	Soft	Up

P → +ve instances
N → -ve instances

Step 1

→ calculating Entropy of class attribute first

$$\text{Entropy}_{\text{class}} = -\frac{5}{10} \log_2\left(\frac{5}{10}\right) - \frac{5}{10} \log_2\left(\frac{5}{10}\right) = 1$$

Step 2: calculate Info Gain, Entropy & Gain for each attribute

$$I[P_i, N_i] = -\frac{P}{P+N} \log_2\left(\frac{P}{P+N}\right) - \frac{N}{P+N} \log_2\left(\frac{N}{P+N}\right)$$

$$\text{Entropy} = \frac{\sum_i P_i + N_i}{P+N} \times I[P_i, N_i]$$

$$\text{Gain} = \text{Entropy}_{\text{class}} - \text{Entropy}_{\text{attribute}}$$

Age

	P_i	N_i	$I(P_i, N_i)$
old	3	0	0
mid	2	2	1
new	3	0	0

N_i, P_i
mein se
koi khee
zero tou
 $I[P_i, N_i] = 0$
same for 1

$$\text{Entropy of age} = \left[\frac{0+3}{10} (0) \right] + \left[\frac{2+2}{10} (1) \right] + \left[\frac{3+0}{10} (0) \right]$$
$$= 0.4$$

$$\text{Gain} = 1 - 0.4 = 0.6$$

Comp

	P_i	N_i	$I[P_i, N_i]$
Yes	1	3	0.81127
NO	4	2	0.918295

$$\text{Entropy of Comp} = \left[\frac{1+3}{10} (0.81127) \right] + \left[\frac{4+2}{10} (0.918295) \right]$$
$$= 0.8754$$

$$\text{Gain} = 1 - 0.8754 = 0.124515$$

Type

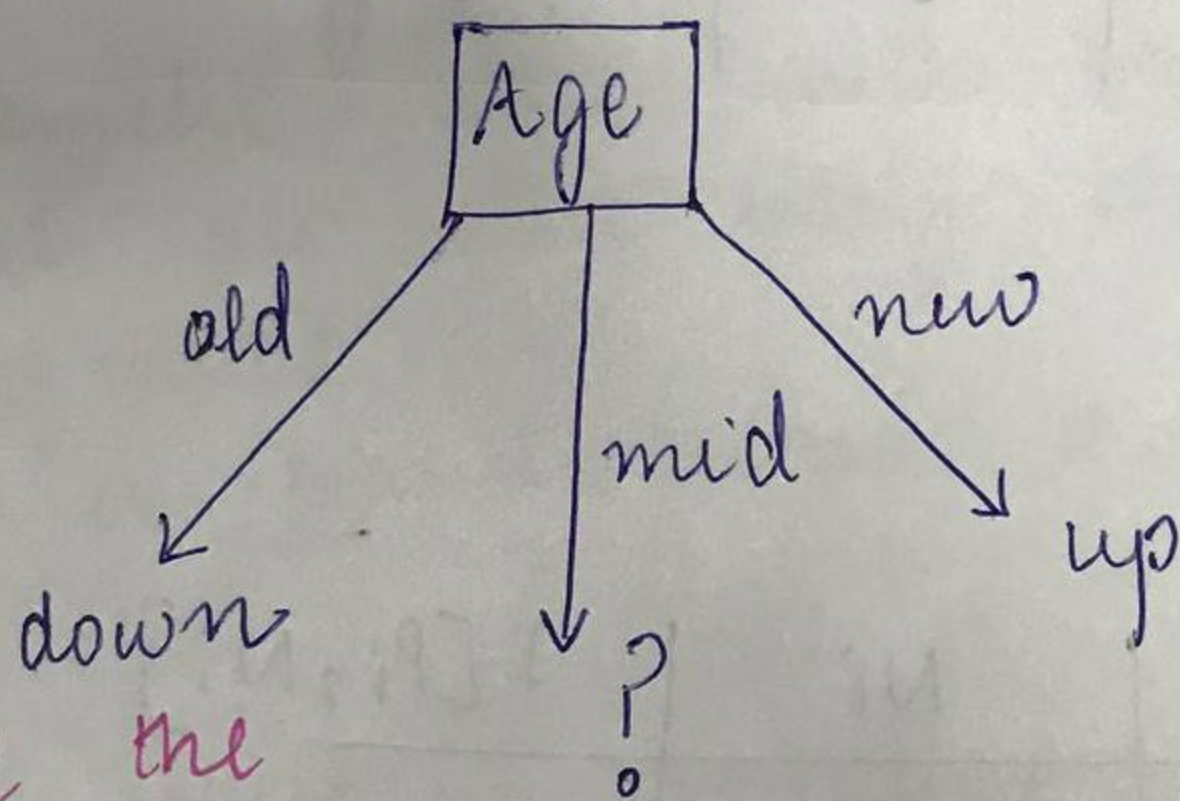
	P_i	N_i	$I[P_i, N_i]$
Soft	3	3	1
Hard	2	2	1

$$\text{Entropy} = \left[\frac{3+3}{10} (1) \right] + \left[\frac{2+2}{10} (1) \right] = 1$$

$$\text{Gain} = 1 - 1 = 0$$

Step 3:

Choose Root! Highest Gain choose!



as all the
value down
there

Recreate Table: with mid only:

~~need~~

T.P.O

Age	Comp	Type	Profit
mid	Yes	Soft	Down
mid	Yes	Hard	Down
mid	No	Hard	Up
mid	No	Soft	Up

$$P = 2$$

$$N = 2$$

$$P + N = 4$$

$$\text{class Entropy} = 1$$

<u>Comp</u>	P_i	N_i	$I[P_i, N_i]$
Yes	0	2	0
No	2	0	0

$$\text{Entropy} = 0$$

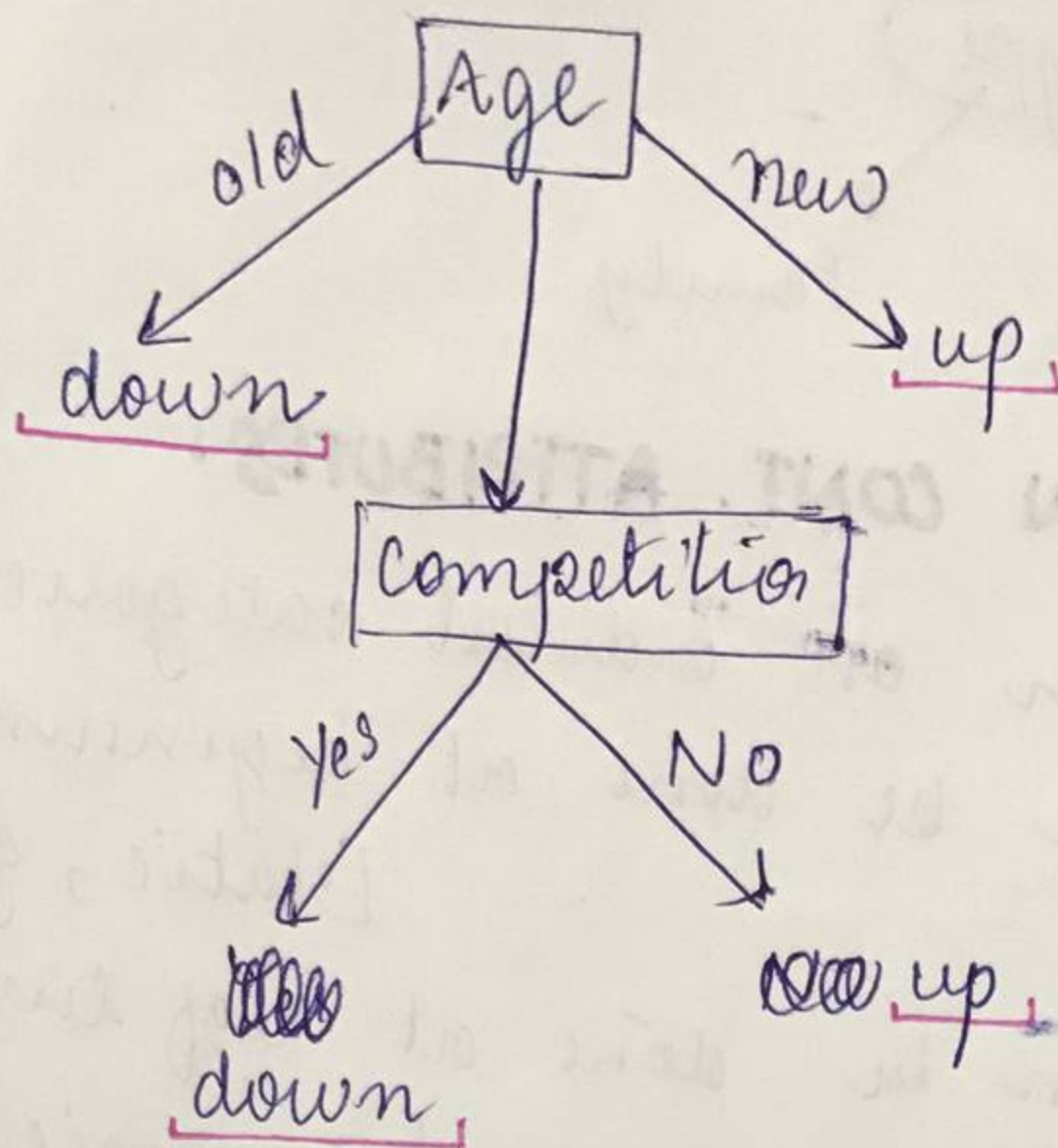
$$\text{Gain} = 1 - 0 = 1$$

<u>Type</u>	P_i	N_i	$I[P_i, N_i]$
Soft	1	1	1
Hard	1	1	1

$$\text{Entropy} = \frac{2}{4}(1) + \frac{2}{4}(1) = 1$$

$$\text{Gain} = 1 - 1 = 0$$

Gain of comp greater



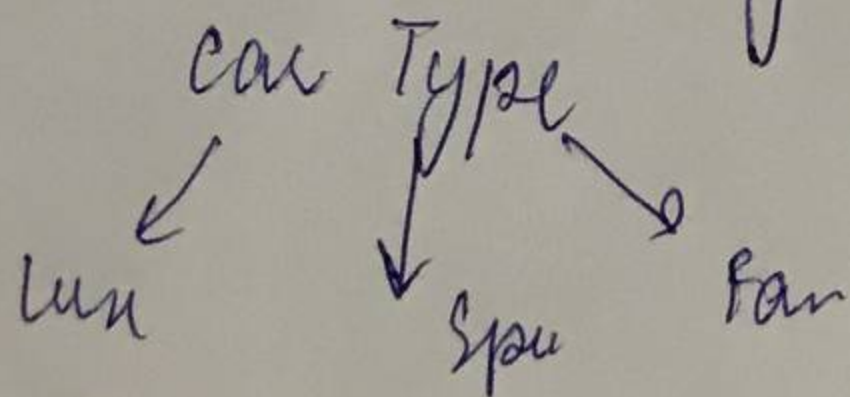
Highly Branching attributes:

- Problematic: attri. with large # of values.
- Subsets are more likely to be pure if there is large # of values
 - I.G is biased towards choosing attributes with large # of values
 - May result in overfitting

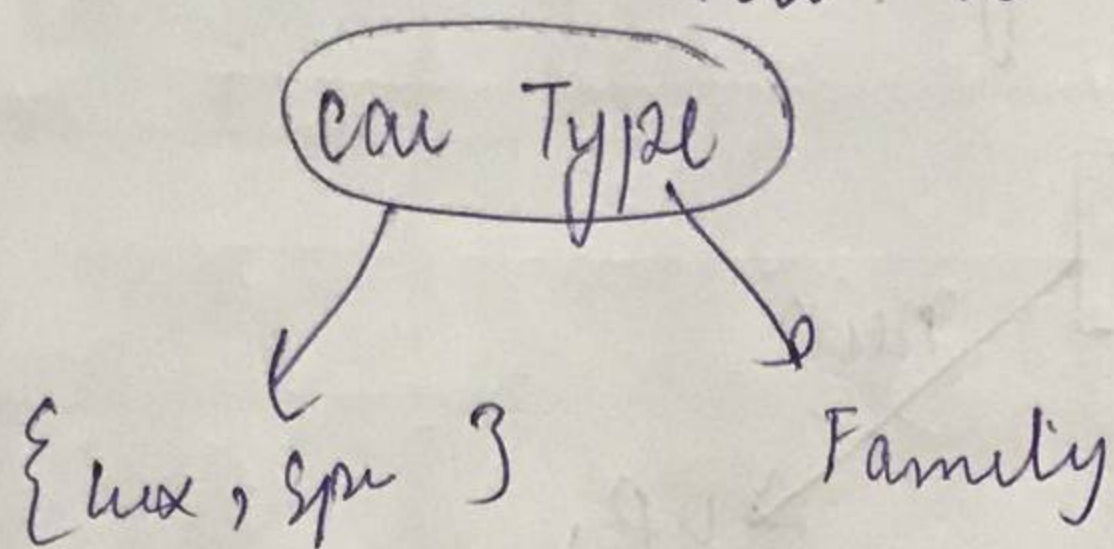
How To specify Test Condition?

- Depends on:
- Type of features/attributes
 - Nominal
 - Continuous
 - Number of ways to split
 - 2-way split
 - Multi-way

Multiway split: use as many partitions as ^{split} there are values.



Binary split Divides into 2 subsets



SPLITTING BASED ON CONT. ATTRIBUTES:

Discretization: → Form an ordinal categorical feature
→ can be done at beginning
[static, global]
→ can be done at leaf level
[dynamic, local]

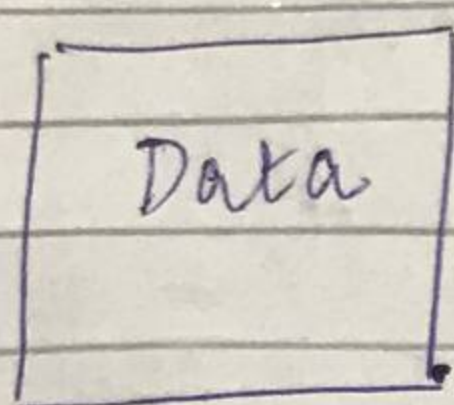
Binary Decision:

- $A < v$ or $A \geq v$
- consider all splits & chooses the best
- More computationally intensive

(Abdullah's, lecture)

PCA - DIMENSIONAL REDUCTION:

⇒ Used for Feature Reduction



(Subset of Data)

Maximum info
must Retain

- $\lambda \rightarrow$ greatest value $\rightarrow$ captures Max var.
- 2nd λ greatest $\rightarrow$ 2nd largest vari. capture
- and so on!

we shall have:

1. Given Data

Step 1

Covariance Find

Step 2

Eigenvalue & $(\lambda_1, \lambda_2, \dots)$

Eigenvector

$$\lambda_1 > \lambda_2 > \lambda_3$$

$$\text{Cov}(x_i, x_j) = \frac{1}{n-1} \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

$$\text{Cov } x, y = \begin{bmatrix} \text{cov}(x, x) & \text{cov}(x, y) \\ \text{cov}(y, x) & \text{cov}(y, y) \end{bmatrix}$$

Note that:

$$\text{cov}(x, y) = \text{cov}(y, x)$$

$$\text{var}(x) = \text{cov}(x, x)$$

Given Data in form of $m \times n$ matrix

$m \rightarrow$ Training Examples

$n \rightarrow$ Features

~~stop~~

$\Rightarrow$ Centralize the data (subtract the mean)

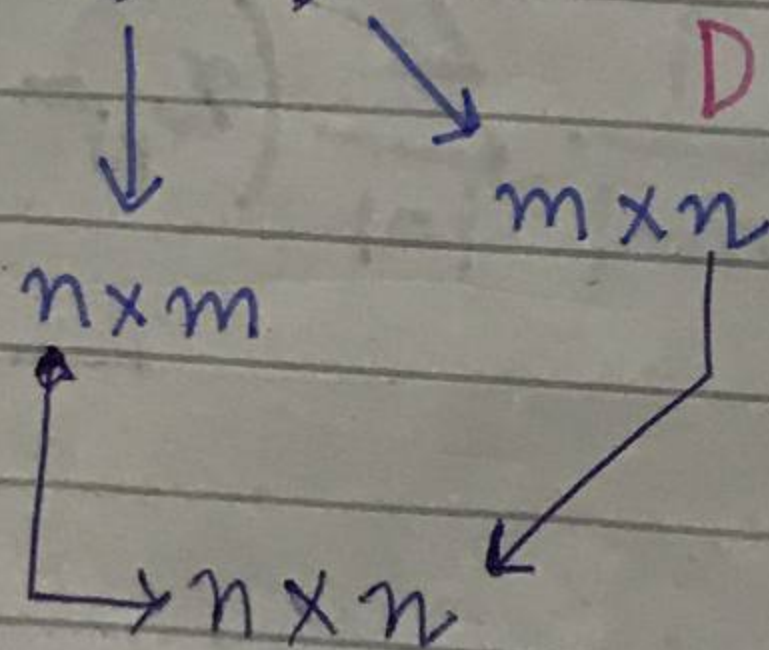
(Method 1)

$$C_{ij} = \text{cov}(d * i, d * j)$$

centralization
not required

(Method 2)

$$C = D^T D \quad \rightarrow \text{1st centralize } D$$



Dim of C

$$C \rightarrow \lambda \rightarrow \Phi$$

Next, $D' = D \Phi \rightarrow$ Eigen Matrix

$\lambda \rightarrow$ Tells kitni Variation

Example:

$$D = \begin{matrix} & x_i & y_i \\ \begin{bmatrix} 2 & 4 \\ 1 & 3 \\ 0 & 1 \\ -1 & 0.5 \end{bmatrix} & 4 \times 2 \end{matrix}$$

$$C = \begin{bmatrix} C_{xx} & C_{xy} \\ C_{yx} & C_{yy} \end{bmatrix} = ?$$

$$\Rightarrow C_{xx} = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$\Rightarrow C_{yy} = \frac{1}{n-1} \sum (y_i - \bar{y})^2$$

$$\Rightarrow C_{xy} = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

x	y	$x_i - \bar{x} = A$	$y_i - \bar{y} = B$	AB	A^2	B^2
2	4	1.5	1.875	2.8125	2.25	3.5156
1	3	0.5	0.875	0.4375	0.25	0.7656
0	1	-0.5	-1.125	0.5625	0.25	1.2656
-1	5	-1.5	-1.625	2.4225	2.25	2.64

$$\bar{x} = \frac{2+1+0-1}{4} = 0.5$$

$$\bar{y} = \frac{4+3+1+5}{4} = 2.125$$

$$\text{Cov}(x, y) = \frac{1}{4-1} \sum AB = \frac{1}{3} (6.23)$$

$$\text{Cov}(y, x) = 2.076$$

$$\text{cov}(x, x) = \frac{1}{3} \sum A^2 = \frac{5}{3} = 1.67$$

$$\text{cov}(y, y) = \frac{1}{3} \sum B^2 = \frac{8.187}{3} = 2.729$$

$$C = \begin{bmatrix} 1.67 & 2.08 \\ 2.08 & 2.73 \end{bmatrix}$$

Step 2: Finding λ Values:

$$|C - \lambda I| = 0$$

$$\left| \begin{bmatrix} 1.67 & 2.08 \\ 2.08 & 2.73 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 1.67 - \lambda & 2.08 \\ 2.08 & 2.73 - \lambda \end{bmatrix} \right| = 0$$

$$(1.67 - \lambda)(2.73 - \lambda) - (2.08)^2 = 0$$

$$4.5591 - 1.67\lambda - 2.73\lambda + \lambda^2 - 4.3264 = 0$$

$$\lambda^2 - 4.4\lambda - \cancel{0.08851} = 0$$

$$0.2331$$

$$\lambda_1 = \cancel{0.0000} \quad 4.346$$

$$\lambda_2 = \cancel{0.0500} \quad 0.0536$$

Formula:

$$CX = \lambda I X$$

$$(C - \lambda I) X = 0$$

$$\begin{bmatrix} 1.67 - 4.346 & 2.08 \\ 2.08 & 2.73 - 4.346 \end{bmatrix} \quad (\text{Plug } \lambda_1) \quad X = 0$$

$$\begin{bmatrix} -2.67 & 2.08 \\ 2.08 & -1.61 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

↓

$$D' = \begin{bmatrix} 2 & 4 \\ 1 & 3 \\ 0 & 1 \\ -1 & 5 \end{bmatrix}_{4 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} \\ \\ \\ \end{bmatrix}_{4 \times 1}$$

$\Rightarrow$ we don't use all lambda's
 takeh Reduction ho
 agar dono use kertaay

$$\Phi = \begin{bmatrix} x_1 & x_3 \\ x_2 & x_4 \end{bmatrix}$$

$$D' = \begin{bmatrix} \\ \\ \\ \end{bmatrix}_{4 \times 2} \quad \text{he ban jani the}$$

Quiz 3:

Q#1: 28×28

$n = 780$ Features

$K = ?$ $\rightarrow$ How many lambda's

Given 99% of variance Retained

$\Rightarrow$ # of lambda's to take is equal to K

$$\frac{1}{m} \sum_{i=1}^m \frac{\|x^i - x^i_{\text{approx}}\|^2}{\|x^i\|^2} \leq 1 - \frac{\% \text{ Var}}{100}$$

$\rightarrow D'$ se

[Abdullah Bhai's lecture #2]

COLLABORATIVE FILTERING WITH KNN:

Users →

	1	2	3	4	5	6	7	8	9	10	11	12
1	1		3		?	5			5		4	
2			5	4			4			2	1	3
3	2	4		1	2		3		4	3	5	
4		2	4		5			4			2	
5			4	3	4	2					2	5
6	1		3		3			2			4	

⇒ Remake Table Using
"Rating - Average"

Avg	Mag	1	2	3	4	5	6	7	8	9	10	11	12
→ 3.6	3.34	-2.6	0	-0.6	0	0	1.4	0	0	1.4	0	0.4	0
→ 3.16	3.29	0	0	1.84	0.84	0	0	0.84	0	0	-1.16	-2.16	0.16
→ 3	3.46	-1	1	0	-2	-1	0	0	0	1	0	2	0
→ 3.4	2.68	0	-1.4	0.6	0	1.6	0	0	0.6	0	0	-1.4	0
→ 3.3	2.70	0	0	0.7	-0.3	0.7	-1.3	0	0	0	0	-1.3	1.7
→ 2.6	2.28	-1.6	0	0.4	0	0.4	0	0	-0.6	0	0	1.4	0

⇒ assign zeros to remaining

$$\text{Sim}(r_1, r_2) = \frac{r_1 \cdot r_2}{|r_1| \times |r_2|}$$

; For the Table
Remade

$$\text{Mag} = \sqrt{(\text{Ent}_1)^2 + (\text{Ent}_2)^2 + \dots}$$

$$\text{Sim}(r_1, r_2) = \frac{(-0.6)(1.84) + ((0.4) \times (-2.16))}{3.34 \times 3.29}$$

$$= -0.179$$

$$\text{Sim}(r_1, r_3) = 0.41$$

$$\text{Sim}(r_1, r_4) = -0.10$$

$$\text{Sim}(r_1, r_5) = -0.31$$

$$\text{Sim}(r_1, r_6) = 0.59$$

Pick highest two neighbours

* Prediction Formula:

$$p_{1,5} = ?$$

movie user

$$p_{i,u} = \frac{\sum_j \text{Sim}_{ij} r_{ju}}{\sum_j \text{Sim}_{ij}}$$

10160

$$p_{1,5} = \frac{(0.41 \times 2) + (0.59 \times 3)}{0.41 + 0.59}$$

$$= 2.59$$

$$= 2.6$$

* USER BY USER :

	M ₁	M ₂	M ₃	M ₄	M ₅	Av	moves →
C ₁	5	3	4	4	?	4	
C ₂	3	1	2	3	3	2.25	
C ₃	4	3	4	3	5	3.5	
C ₄	3	3	1	5	4	3	
C ₅	1	5	5	2	1	3.5	

• Average without using column 5 :

Remake Table :
(Rating - Avg)

	M ₁	M ₂	M ₃	M ₄	M ₅	Mag Ex 5	Mag In 5
C ₁	1	-1	0	0	?	1.41	-
C ₂	0.75	-1.25	-0.25	0.75	0.75	1.658	1.82
C ₃	0.5	-0.5	0.5	-0.5	1.5	1	1.80
C ₄	0	0	-2	2	1	2.82	3
C ₅	-2.5	1.5	1.5	-1.5	-2.5	3.605	4.387

$$\text{Sim}(C_1, C_2) = \frac{C_1 \cdot C_2}{|C_1| \times |C_2|} = \frac{(0.75 \times 1) + (-1 \times -1.25)}{1.41 \times 1.658}$$

$$= 0.8529$$

$$\text{Sim}(C_1, C_3) = 0.7$$

$$\text{Sim}(C_1, C_4) = 0$$

~~$$\text{Sim}(C_1, C_5) = 0$$~~

$$\text{Sim}(C_1, C_5) = -0.79$$

$$\beta_{i,u} = ?$$

Ayesha's
chat ques!

$$y = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

$$\theta_0 = 6$$

$$\theta_1 = 0$$

$$\theta_2 = -1$$

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 = z$$

$$z < 0 \text{ For } y = 0$$

$$z \geq 0 \text{ For } y = 1$$

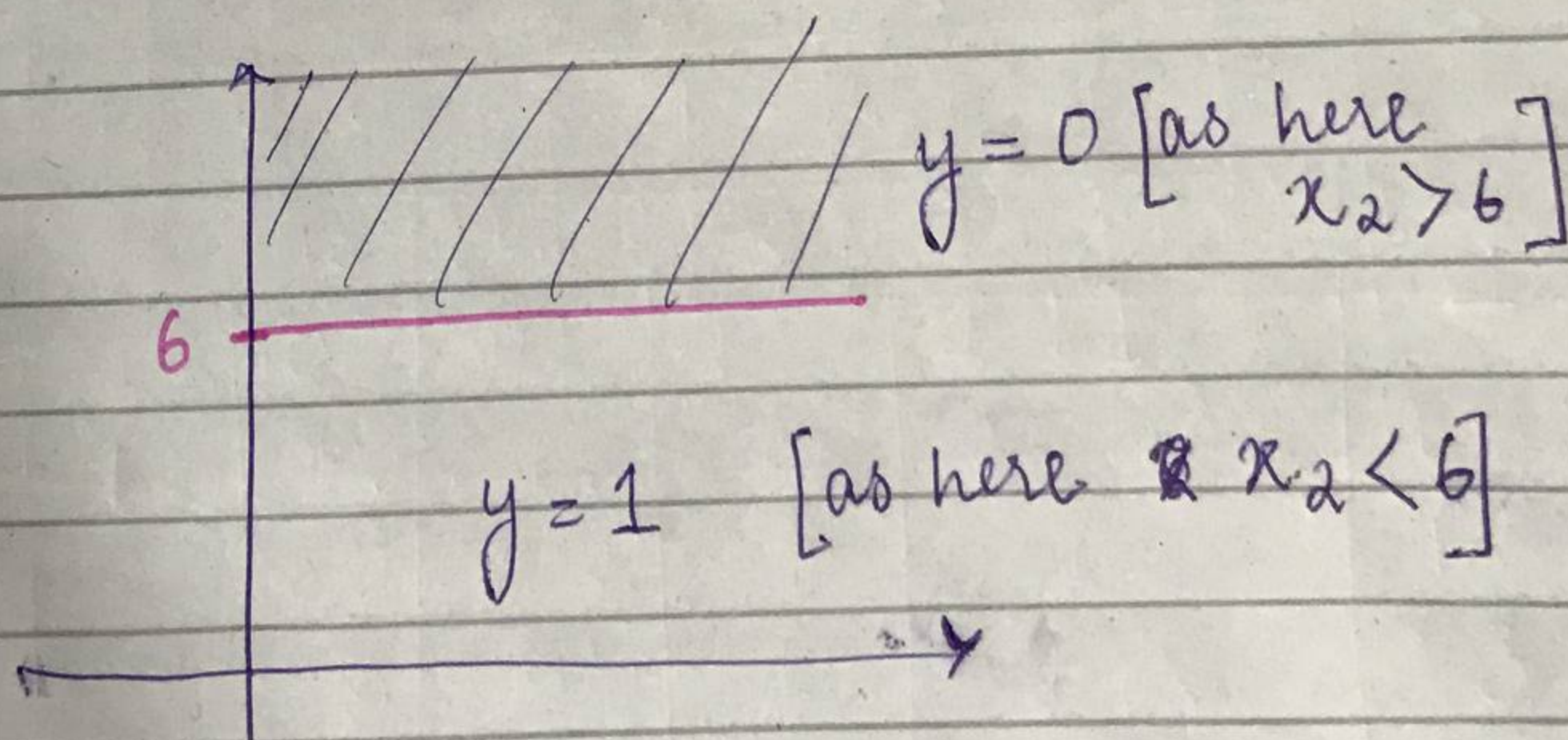
$$z < 0$$

$$6 - x_2 < 0$$

$$-x_2 < -6$$

$$x_2 > +6$$

condition for $y = 0$



Derivative term gets smaller and smaller and also at local minimum -

PYTHON IMPLEMENTATION OF LINEAR REGRESSION USING

Q.D - see Assig

①

1.1: Machine Learning:

Field of study that gives computers the ability to learn without being explicitly programmed.

A computer program is said to learn from experience E w.r.t some Task T and some performance measure P , if its performance on T , as measured by P ,

improves with E

Supervised Learning

Unsupervised Learning

ML algorithms

Reinforcement Learning

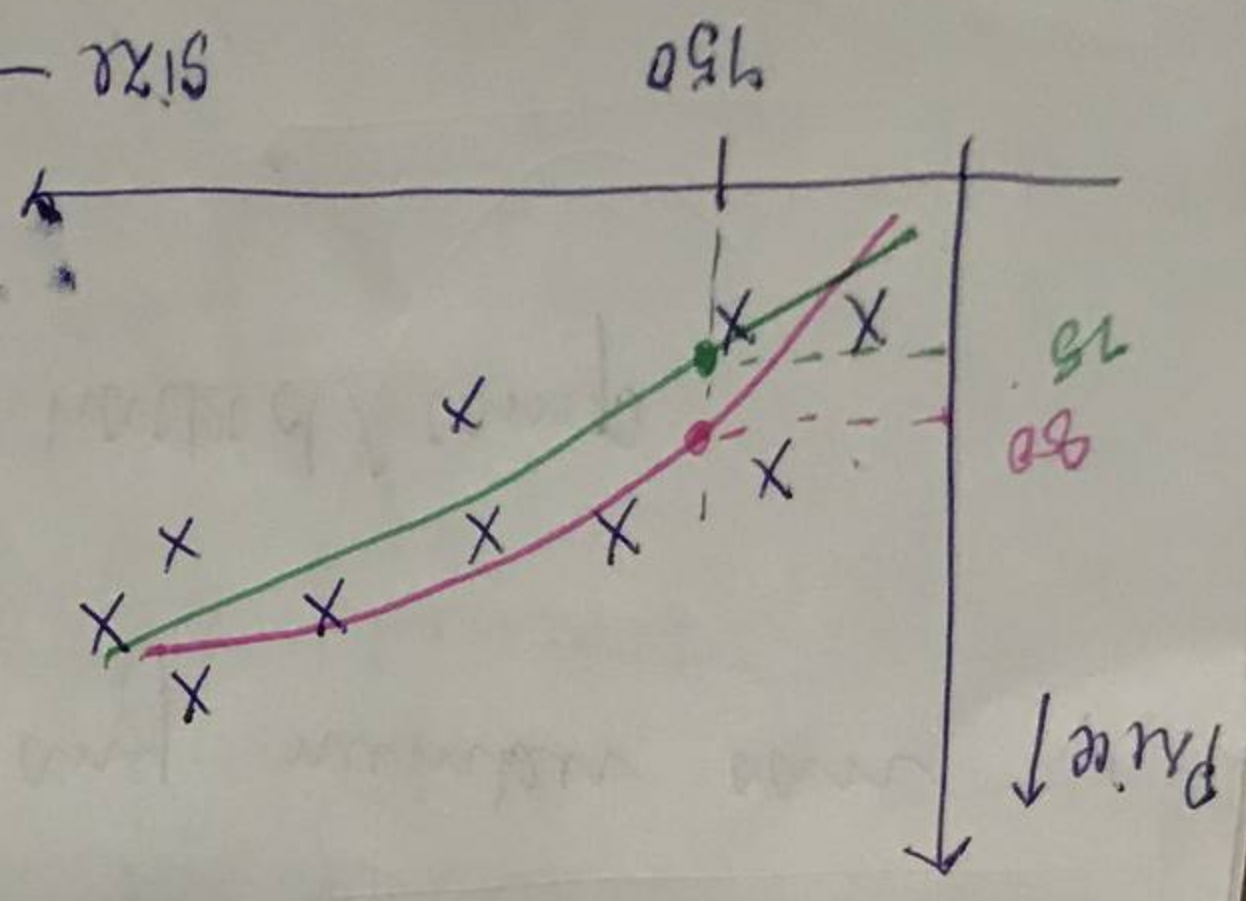
Recommendation Systems

Others

MOST IMPORTANT ALGO'S

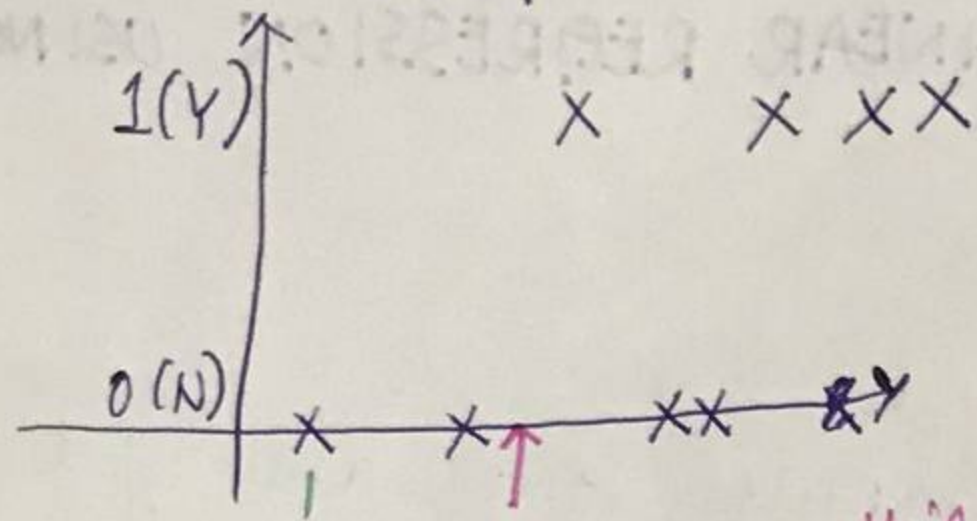
1.2 SUPERVISED LEARNING:

- We have this data
- Friend wants 750 size house for model
- 80k price -
- This is supervised learn algorithm as:



S.L → Right answers given

we model to get ② more of these right.
 $\Rightarrow$ Regression: Predict continuous valued output
 (price) mal ↑



$\Rightarrow$ CLASSIFICATION:

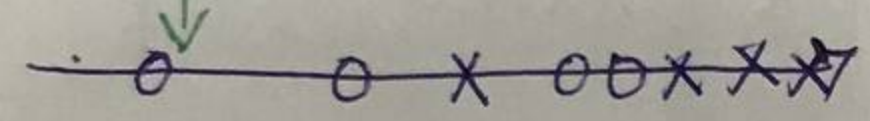
Discrete valued output

unsupervised
 unlabeled
 clustering
 application
 organi
 ization
 ra

Friend has this sized Tumor

predict if mal?

Another way of mapping:



T. size

$\Rightarrow$ labeled Data Set Given

Prob: large inventory of iden. items - Predict how many will sell over the next 3 months — ①

You'd like software to examine individual customer accounts, and for each account decide if it has been hacked / compromised — ②

① Regression - 1000's so any number can sell -

② classification - as either hacked / comp
 output discrete prob

unsupervised learning :

③

unlabeled Dataset given

clustering Algorithm: [Google News e.g]

Applications :

Organize computing clusters

social network analysis

→ identify cohesive grp
of ppl who know each other

Market segmentation → as we don't know in advance
who's in what segment

Astronomical data analysis.

→ How galaxies formed & baray mein interesting
information

2.1 Linear Regression :

→ Predict Real Valued Output

Notation :

m → Number of Training Examples

x 's → Input variable / Features

y 's → Output variable / 'Target' variable

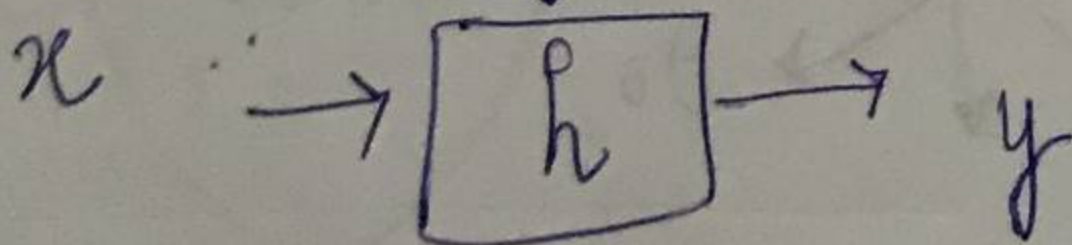
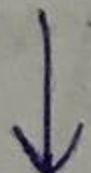
(x^i, y^i) i th training e.g

ALGO:

Training Set



Learning Algo



$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

COST FUNCTION: [2.2]

θ_1 's $\rightarrow$ Parameters

How to choose parameters

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left[(h_{\theta}(x^{(i)}) - y^{(i)})^2 \right]$$

Minimize this

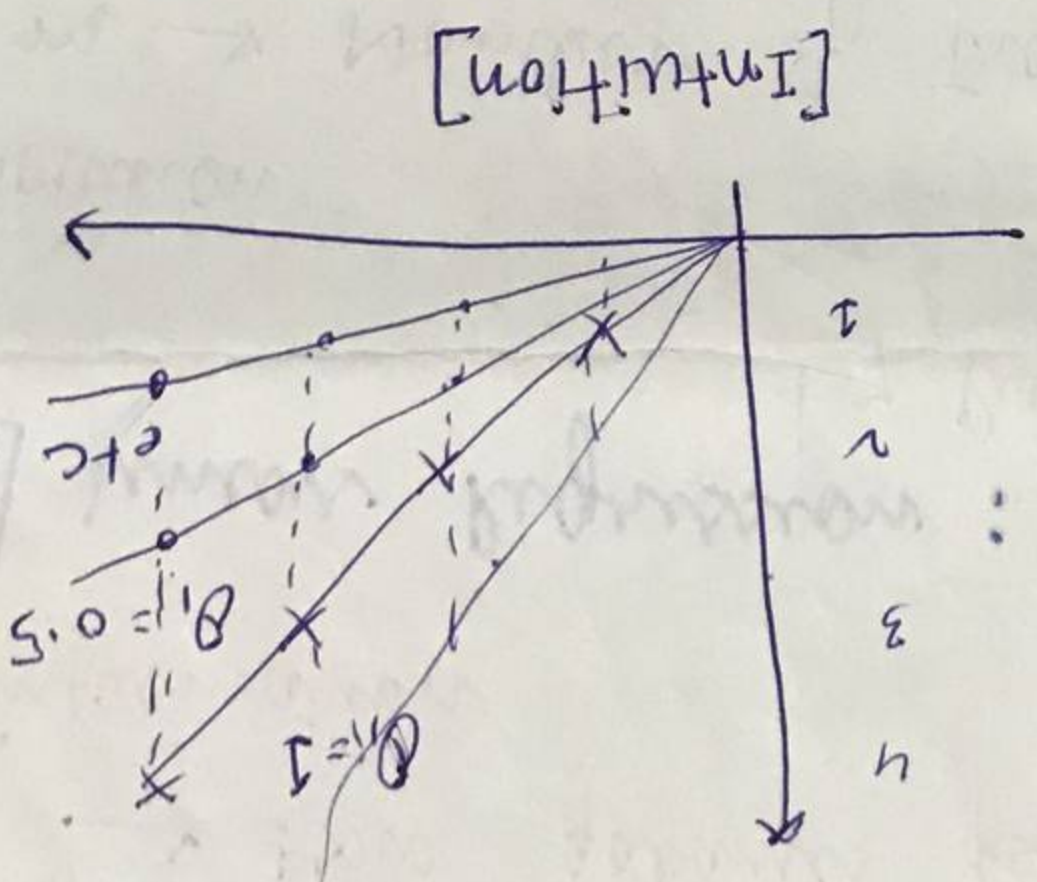
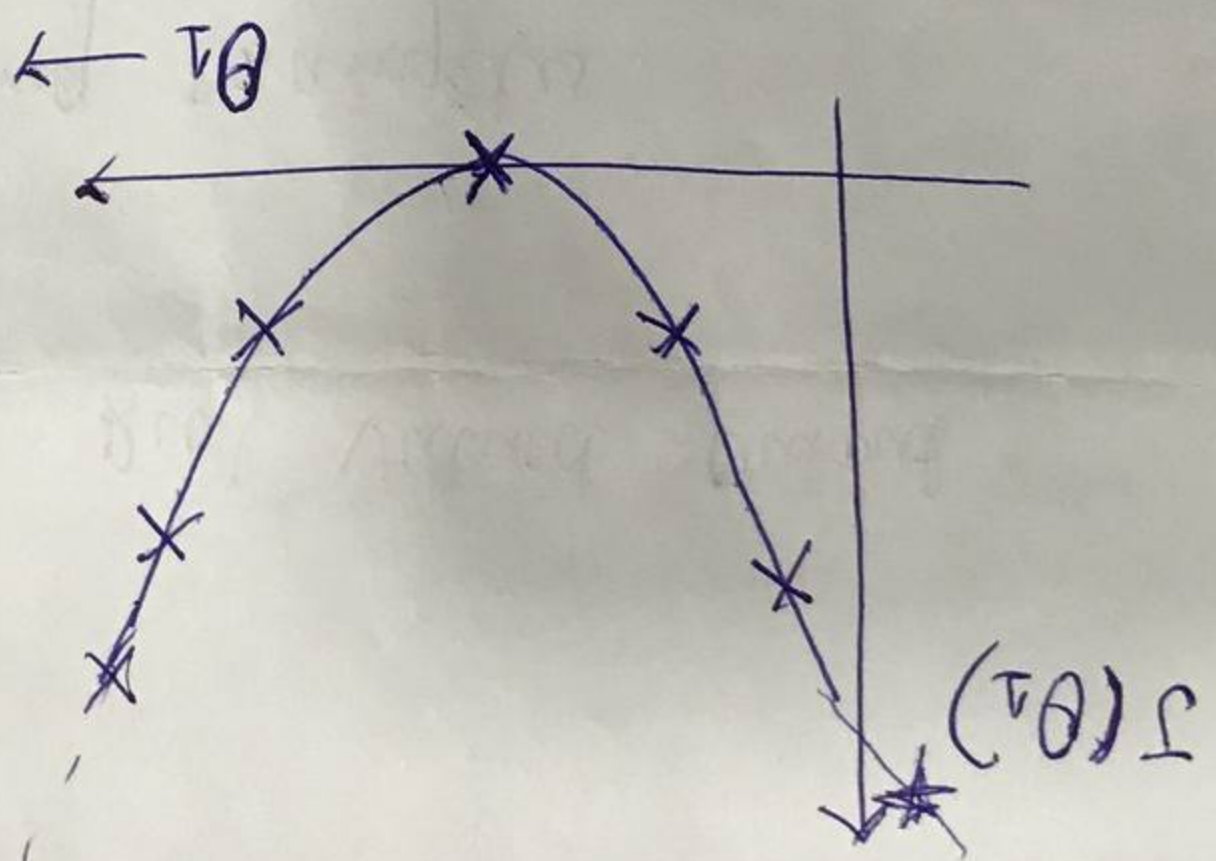
by choosing θ_0, θ_1

$\rightarrow$ Squared Error Function

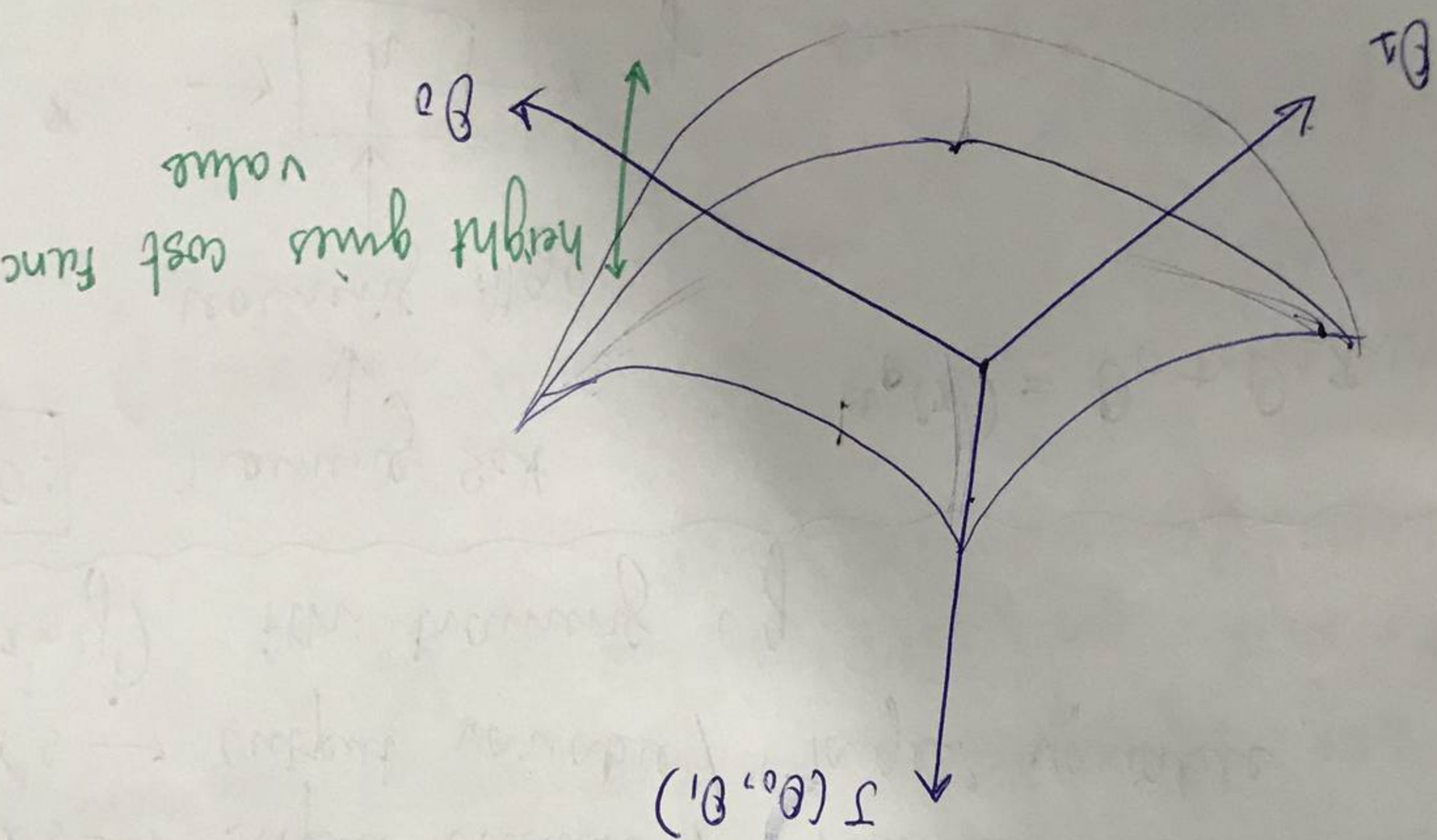
2.3

if $\theta_0 = 0$ for simplicity

$J(\theta_1)$



2.4



Linear Regression with One Variable - Gradient Descent

G.D used to minimize cost function here. [ML]

$$J(\theta_0, \theta_1) ; \text{ want } \min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$$

2.5

5

Outline:

- Start with some θ_0, θ_1 (any) → Depending upon it, diff local minima. pe
- Keep changing θ_0, θ_1 to reduce $J(\theta_0, \theta_1)$ converge

GRADIENT DESCENT ALGORITHM:

Repeat until convergence

$$\{ \theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) \}$$

assignment operator

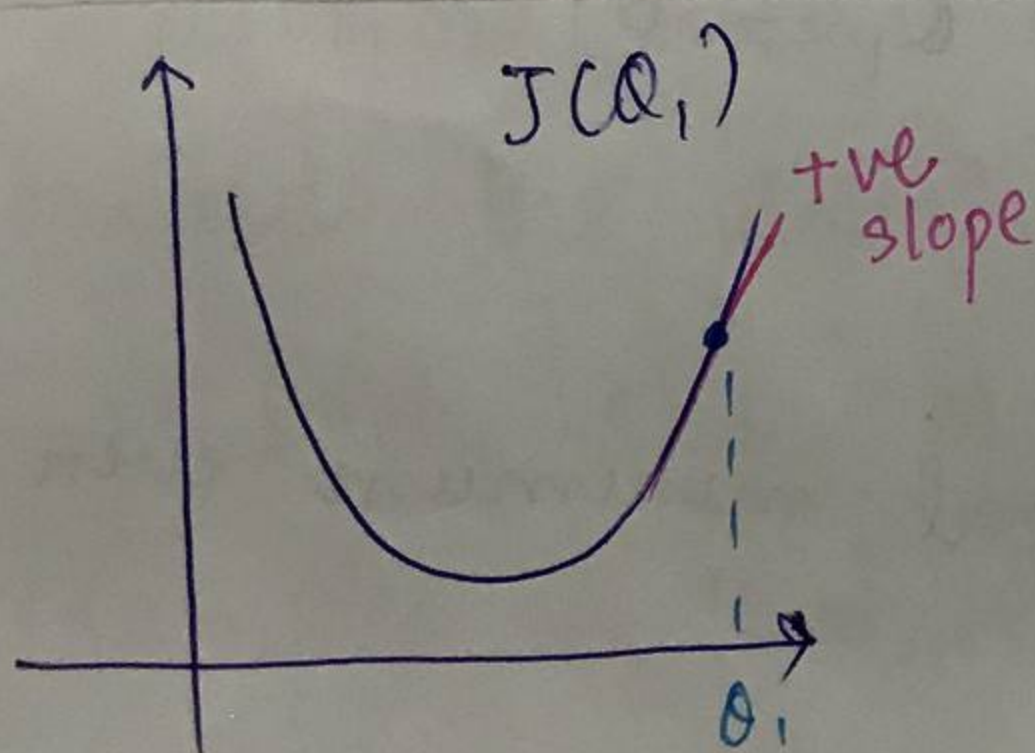
learning rate

How big a step we take

for $j=0, j=1$

• update simultaneously

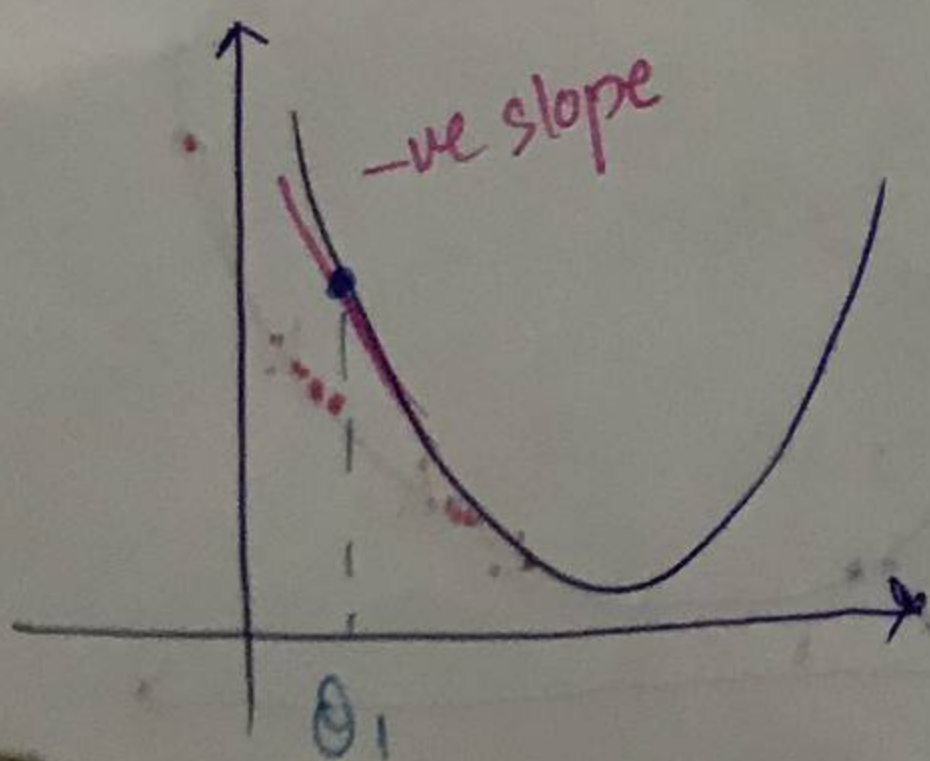
2.6



$$\bullet \theta_1 := \theta_1 - \alpha \frac{d}{d\theta_1} J(\theta_1)$$

$$\theta_1 := \theta_1 - \alpha (\text{+ve number})$$

→ decrease θ_1



$$\bullet \theta_1 := \theta_1 - \alpha (-\text{ve \#})$$

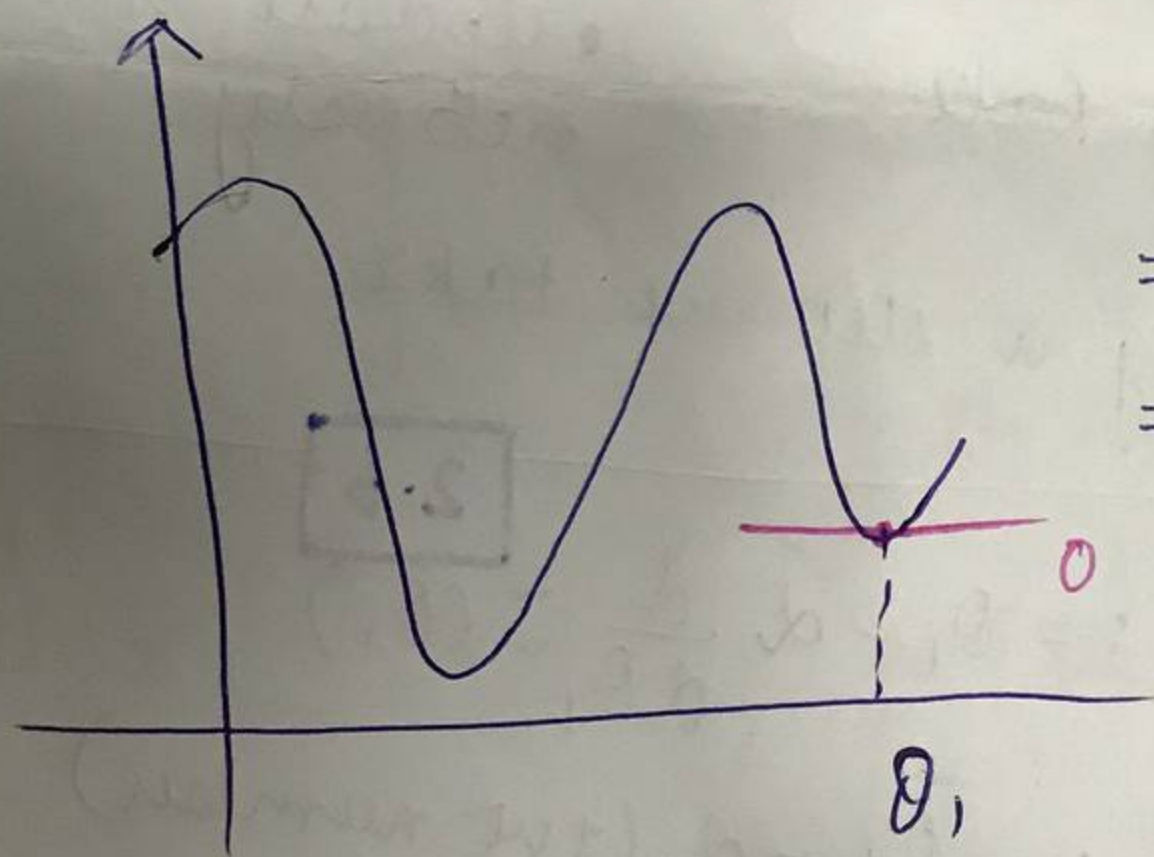
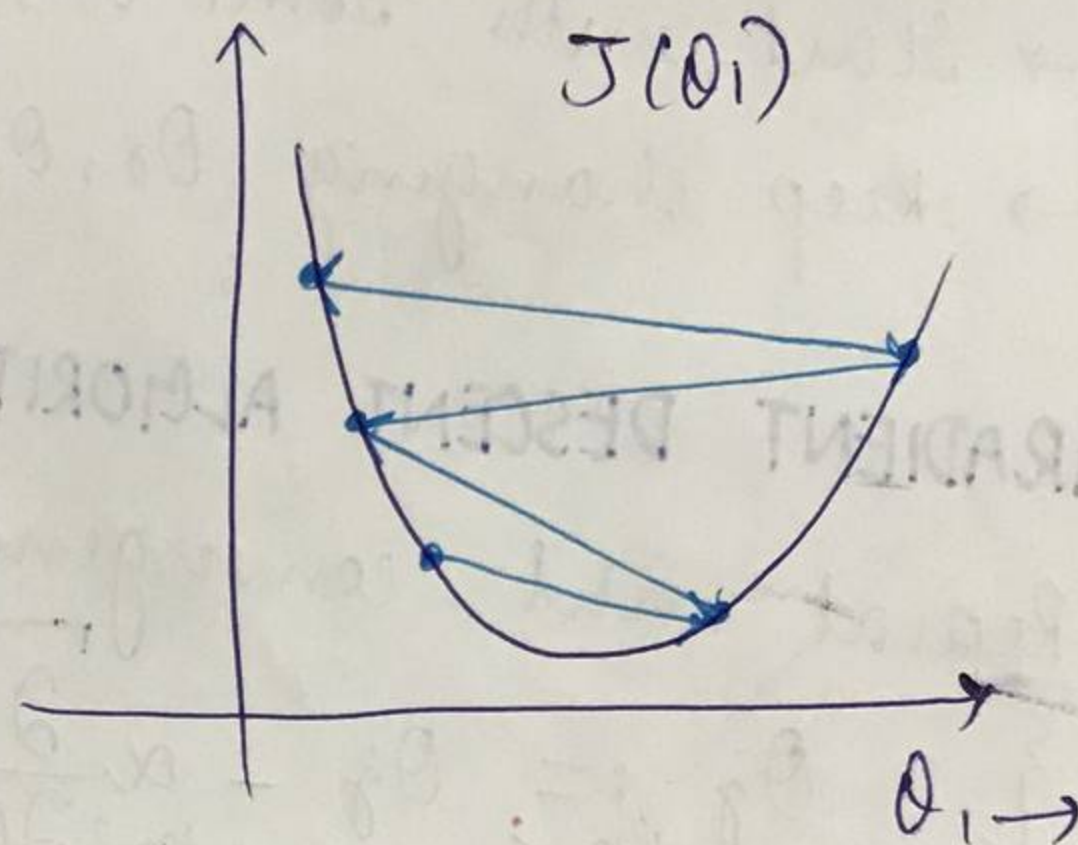
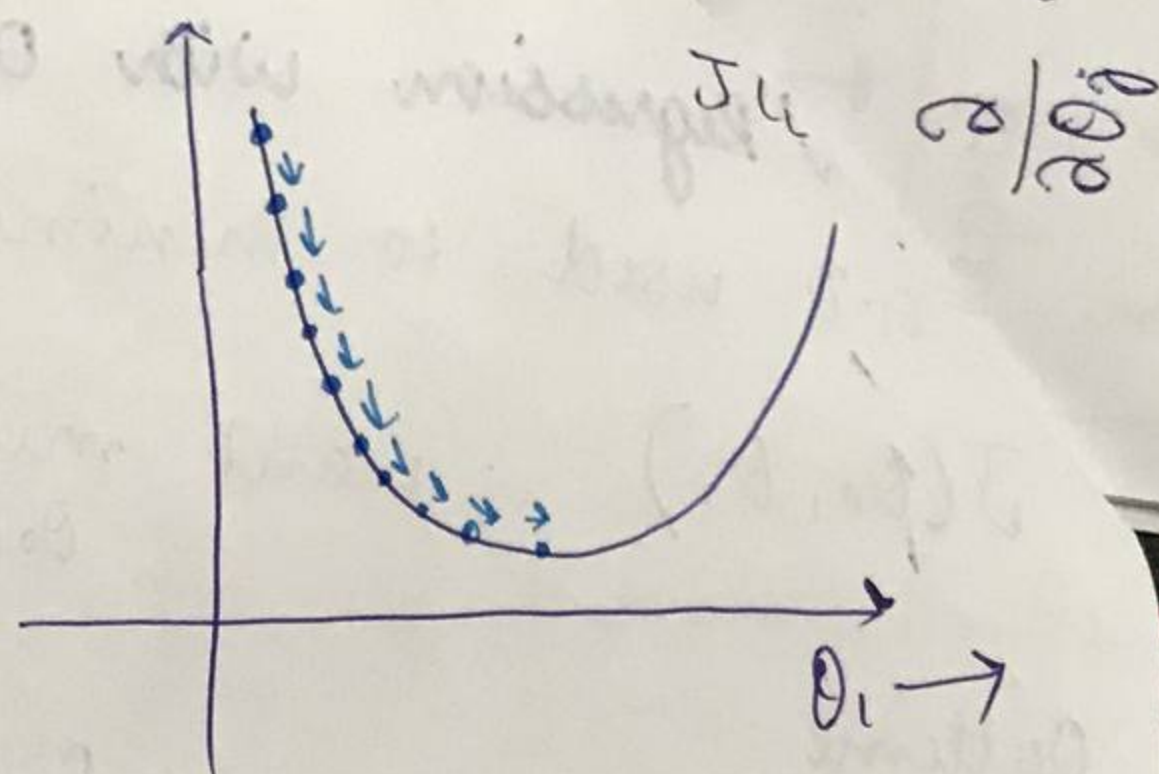
→ Increase θ_1

α :
Too Small :

→ Gradient Descent
can be ~~small~~ slow

α :
Too Large :

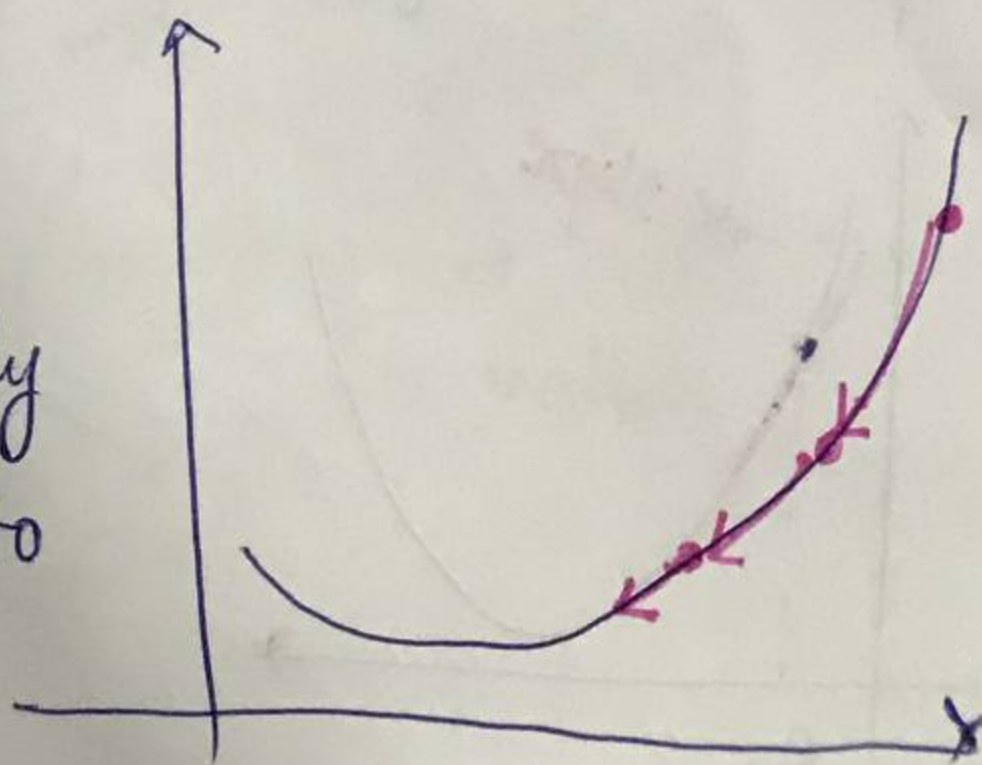
→ Can overshoot the Minimum
→ Fail to converge or even
diverge



$\Rightarrow$ If θ_1 at local minima
 $\Rightarrow$ as $\theta_1 := \theta_1 - \alpha [0]$
 $\theta_1 := \theta_1$

• G.D can converge to a local minimum even
with α fixed

$\Rightarrow$ As we approach local
minimum, G.D will automatically
take smaller steps, so no need to
decrease α over time.



$$\frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1) = \frac{\partial}{\partial \theta_j} \cdot \frac{1}{2m} \sum_{i=1}^m [h_0(x)^{(i)} - y^{(i)}]^2$$

$$= \frac{\partial}{\partial \theta_j} \frac{1}{2m} \sum_{i=1}^m [\theta_0 + \theta_1 x^{(i)} - y^{(i)}]^2$$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{m} \sum (h_0(x)^{(i)} - y^{(i)})$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{m} \sum [h_0(x)^{(i)} - y^{(i)}] x^{(i)}$$

REPEAT
UNTIL
CONVERGENCE.

UPDATE
SIMULTANEOUSLY

In Linear Regression:

$J(\theta_0, \theta_1)$ nota tou convergence is always towards the global Minima

$\Rightarrow$ Each step of G.D uses all of the Training examples - "Batch"

$$(Q+2)^2 = Q^2 + 4Q + 4$$

4.1 Multiple Features

(8)

L.R

Notation:

$n \rightarrow \#$ of Features

$m \rightarrow \#$ of Training Examples

$x_j^{(i)}$ $\rightarrow$ i th Training Example's value for j^{th} Feature

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \dots + \theta_n x_n$$

$$X = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_n \end{bmatrix}$$

$$x_0 = 1$$

$$h_{\theta}(x) = \theta^T X$$

Multiplication condition - CR

4.2

G.D

Cost Func

$$J(\theta_0, \theta_1, \dots, \theta_n) = \frac{1}{2m} \sum_{i=1}^m [h_{\theta}(x^{(i)}) - y^{(i)}]^2$$

$\hookrightarrow$ call it $J(\theta)$

Repeat until convergence:

$$n = 1$$

$$\left\{ \begin{aligned} \theta_j &:= \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) \end{aligned} \right.$$

}

1:

(9)

repeat

$$\theta_j := \theta_j - \alpha \frac{1}{m} \left[\sum_{i=1}^m (h_0(x)^i - y^{(i)}) x_j^{(i)} \right]$$

Simultaneously update θ_j for $j = 0, \dots, n$

4.3 Feature Scaling

→ Make sure Features are on a similar scale

Get every feature into approximately $-1 \leq x_i \leq 1$ range

• Mean Normalization

Replace x_i with $x_i - \mu_i$ to make features have approximately zero mean (Do not apply to $x_0=1$)

$$\frac{x_1 - \mu_1}{s_1}$$

Average Value / Mean

Range MaxVal - Min Value

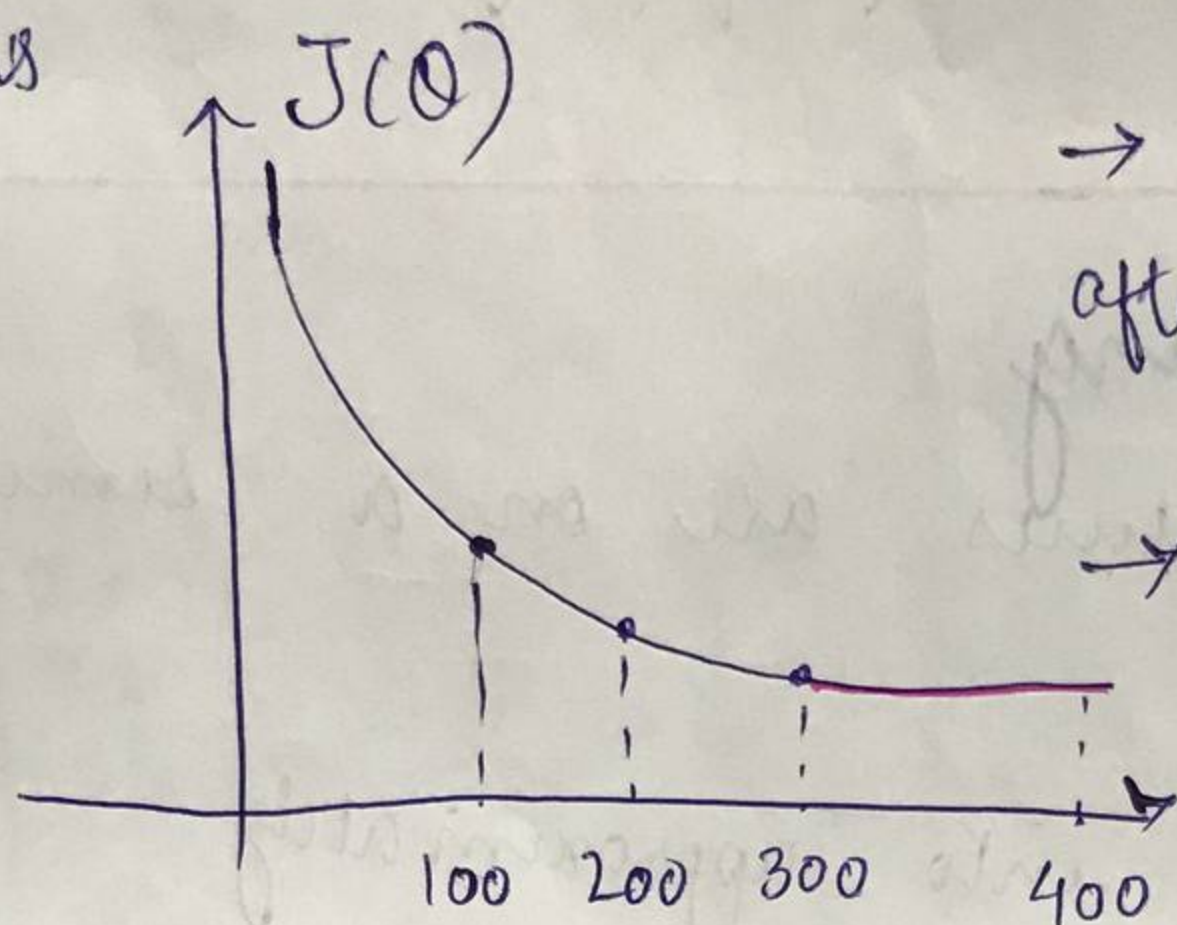
or
Stand. Dev

4.4

→ Choosing α

→ Making sure G.D working

① Plot the cost Function with number of iterations



→ $J(\theta)$ should decrease after every iteration

→ when curve flattens G.D has converged

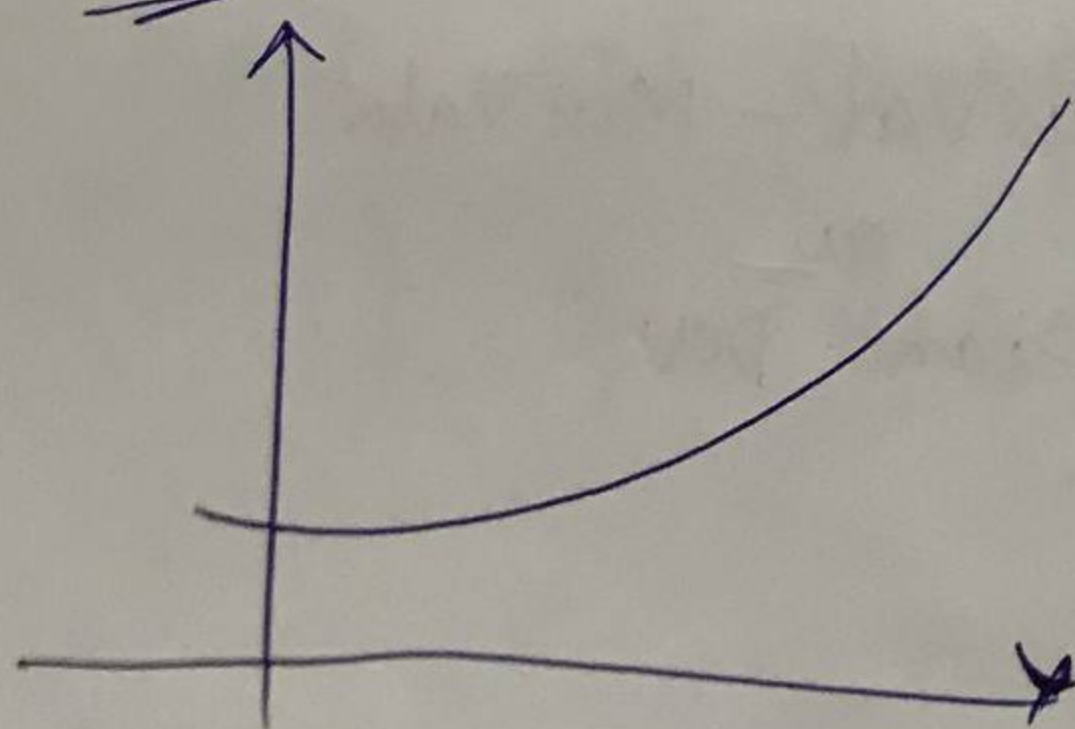
- Different App's k take diff iterations to converge.

- You App k accord. change/tune

→ Declare convergence if $J(\theta)$ decreases by less than 10^{-3} in one iteration.

(For Automatic)

CASE

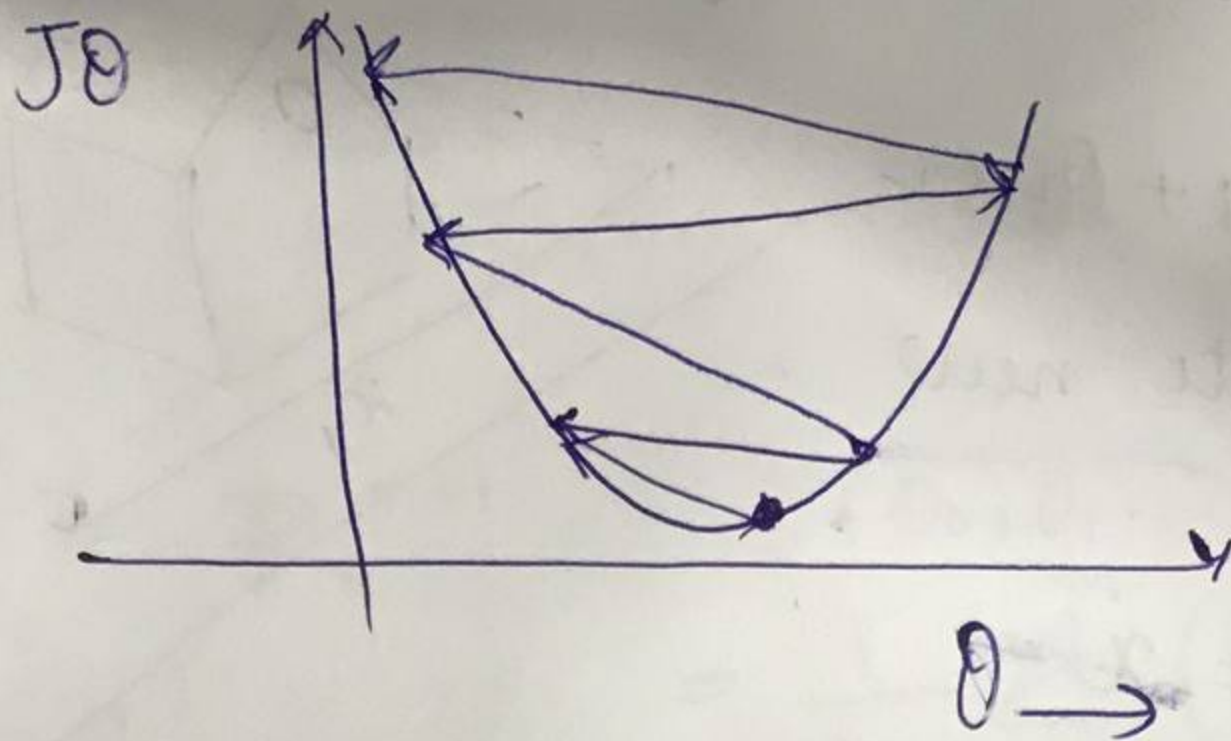


→ G.D not working

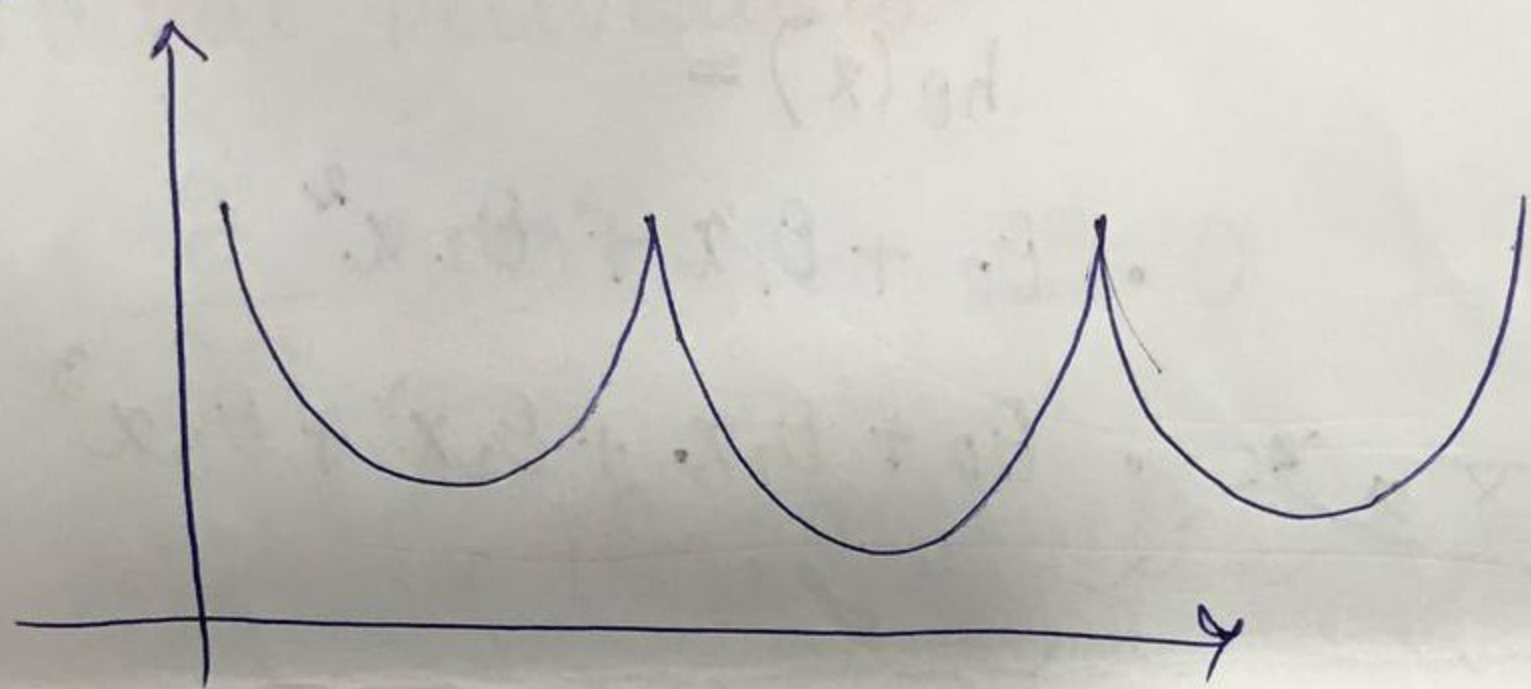
→ Use smaller α

kerla yeh

(11)



CASE :



Fix $\rightarrow$ Use smaller α

Summary :

$\rightarrow \alpha$ small $\rightarrow$ slow convergence

$\rightarrow \alpha$ large $\rightarrow$ may not decrease on every iteration
; may not converge ; may even diverge

NORMAL EQUATION:

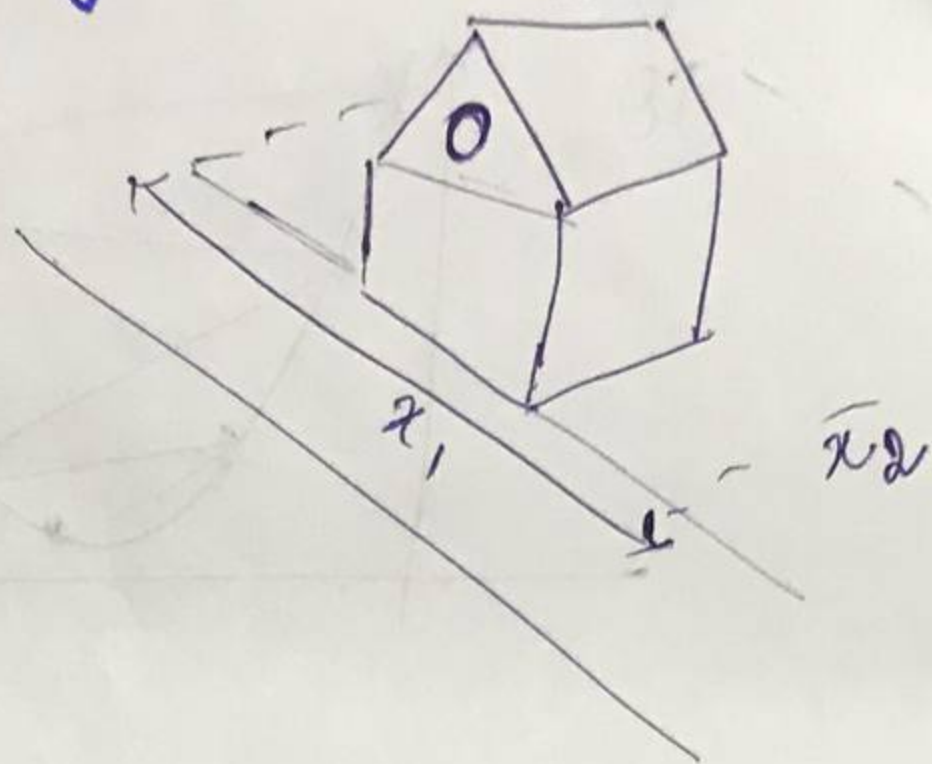
4.4

Features & Polynomial Regression: 45

$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

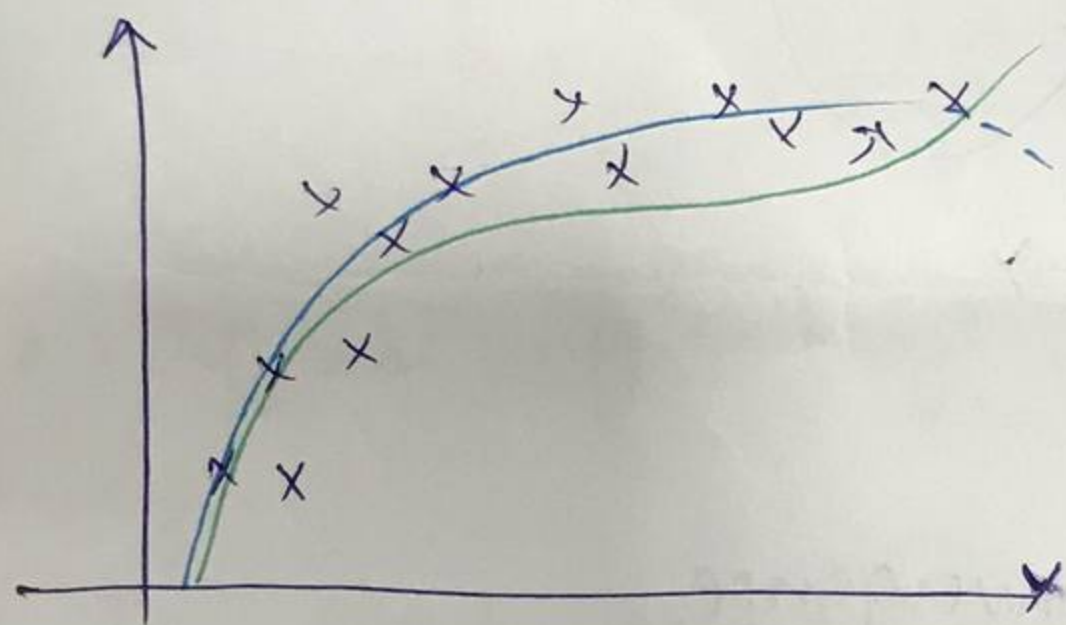
→ we can create new features:

$$\text{Area} = x_1 x_2 = x$$



$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Polynomial Regression:



$$h_{\theta}(x) =$$

$$\bullet \theta_0 + \theta_1 x + \theta_2 x^2$$

$$\bullet \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

to compensate for --- we might model

$$h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 \sqrt{x}$$

4.6 NORMAL EQUATION:

Method to solve for θ analytically

$$J(\theta) = a\theta^2 + b\theta + c \quad (13)$$

$$\frac{d}{d\theta} J(\theta) = \dots \stackrel{\text{set}}{=} 0 \quad \text{solve for } \theta$$

But $\theta \in \mathbb{R}^{n+1}$

[Minima Find]

$$J(\theta_0, \theta_1, \dots, \theta_m) = \frac{1}{2m} \sum_{i=1}^m (h_0(x)^{(i)} - y^{(i)})^2$$

Partial Derivatives:

$$\frac{\partial}{\partial \theta_j} J(\theta) = \dots \stackrel{\text{set}}{=} 0 \quad \text{for every } j$$

solve for $\theta_0, \theta_1, \dots, \theta_n$

E.g.:

x_1	x_2	x_3	x_4	y	x_0
2104	5	1	45	460	1
1416	3	2	40	232	1
1534	3	2	30	315	1
852	2	1	36	178	1

$$X = \begin{bmatrix} 1 & 2104 & 5 & 1 & 45 \\ 1 & 1416 & 3 & 2 & 40 \\ 1 & 1534 & 3 & 2 & 30 \\ 1 & 852 & 2 & 1 & 36 \end{bmatrix}$$

$m \times (n+1)$

$$Y = \begin{bmatrix} 460 \\ 232 \\ 315 \\ 178 \end{bmatrix}$$

$m \times 1$

$$\theta = (X^T X)^{-1} X^T Y \quad (14)$$

General:

m examples

n features

⇒ RC way of order writing

$$x^{(i)} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix}$$

→ So Design Matrix

$$X = \begin{bmatrix} \leftarrow (x^{(1)})^T \rightarrow \\ \leftarrow (x^{(2)})^T \rightarrow \\ \vdots \\ \leftarrow (x^{(m)})^T \rightarrow \end{bmatrix}$$

$m \times (n+1)$

$$Y = \begin{bmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(m)} \end{bmatrix}$$

⇒ No need of feature Scaling

G.D

- Need to choose α
- Many iterations
- Feature scaling
- Works well with large n

→ Logistic k lye not good

→ No α

→ No iter

→ No Feature Scal

→ Need to compute $(X^T X)^{-1}$

→ Slow if n large

cost $O(n^3)$

15

Non-Invertibility

- Redundant features $x_1 = \text{size in feet}$
 $x_2 = \text{ " " m}^2$

- Too many features $[m \leq n]$

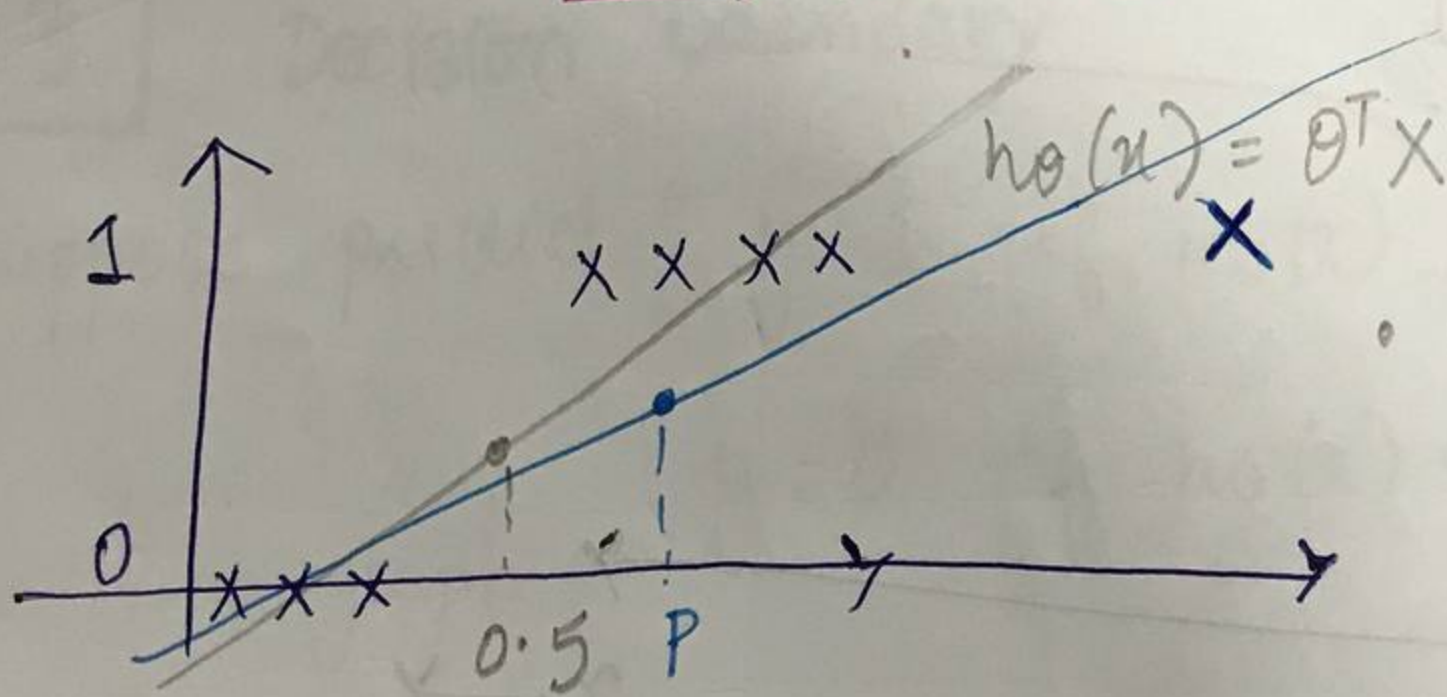
→ Delete some features or use regularization

6.1 Logistic Regression [classification probs]

- e.g's
- Email (Spam / Not Spam)
 - Tumor (Mal / Benign)

$$y \in \{0, 1\}$$

→ -ve class (for 0)
→ +ve class (for 1)



- we could threshold
 $h_0(x) \geq 0.5 \quad y=1$
 $h_0(x) < 0.5 \quad y=0$

⇒ Bad (due to this X) ~~h0(x)~~ as threshold

P gets very wrong - so LR application not a good choice here. usually to a classification problem

16
Lo.R $\rightarrow$ classification algorithm

$$0 \leq h_{\theta}(x) \leq 1$$

Lo.R $\rightarrow$ logistic Reg

L.R $\rightarrow$ linear Reg

In L.R $h_{\theta}(x) < 0$

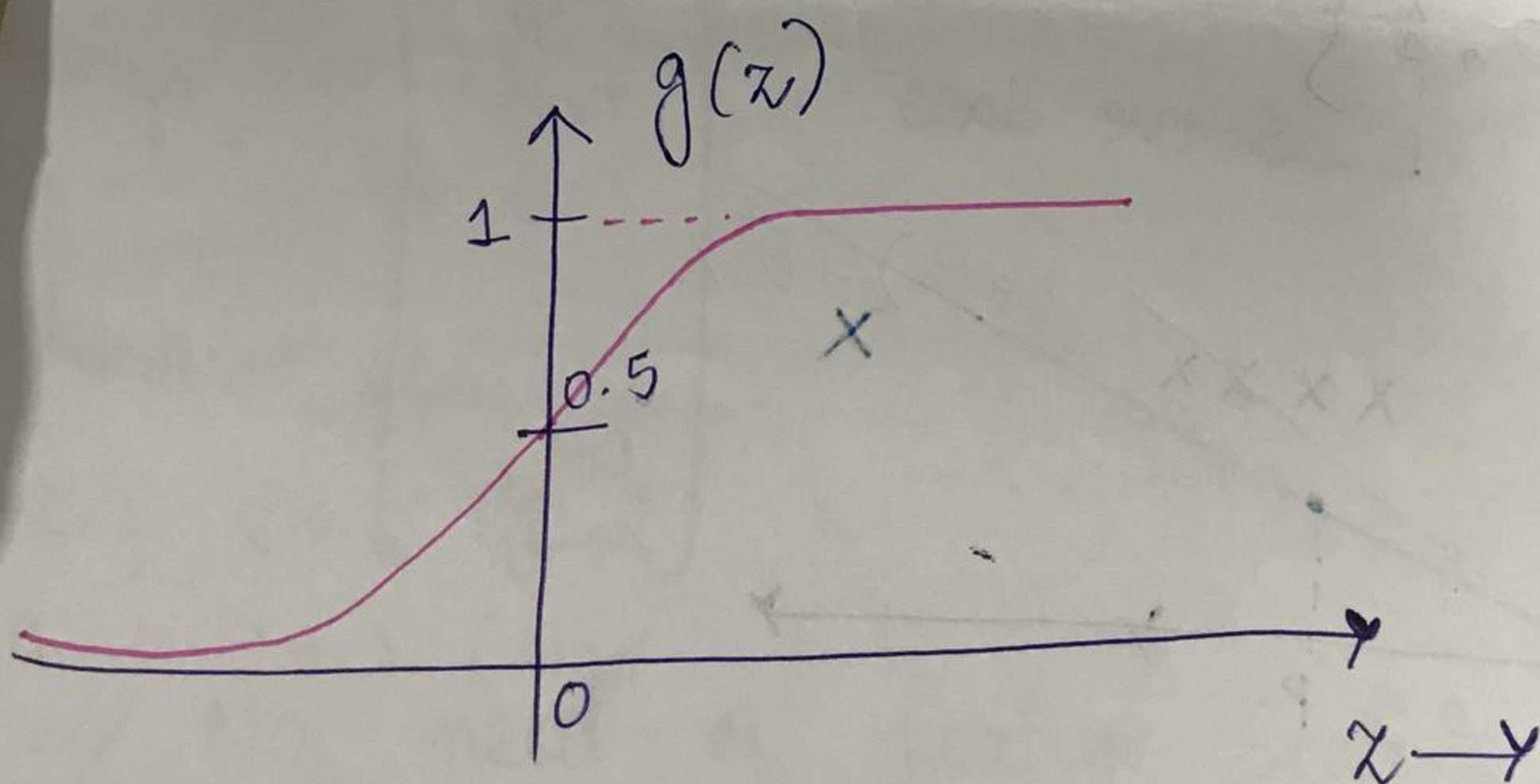
$h_{\theta}(x) > 1$

6.2 Hypothesis Representation:

$$h_{\theta}(x) = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}}$$

Sigmoid function
logistic function



• Interpretation of Hypothesis Output

$h_{\theta}(x)$ = estimated probability that $y=1$ on input x

$$X = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 \\ \text{tumor size} \end{bmatrix} \quad (17)$$

$$h_\theta(x) = 0.7$$

Tell patient has 70% chance of tumor being malignant

$$h_\theta(x) = P(Y=1 | x; \theta) \quad \text{"probability that } y=1, \text{ given } x, \text{ parametrized by } \theta \text{"}$$

$$P(Y=0 | x; \theta) = 1 - h_\theta(x)$$

6.3 Decision Boundary

suppose predict " $y=1$ if $h_\theta(x) \geq 0.5$ "

" $y=0$ if $h_\theta(x) < 0.5$ "

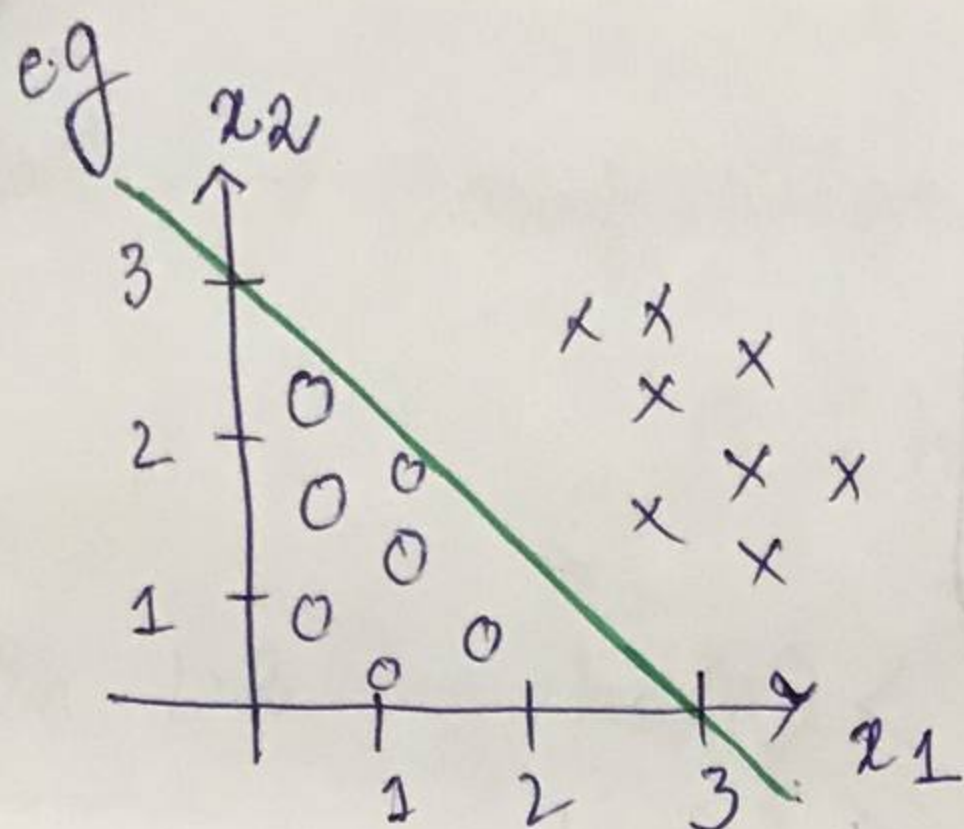
• See Prev graph:

① $g(z) \geq 0.5$ when $z \geq 0$ ~~when~~

② $g(z) < 0.5$ when $z < 0$

also ① when $\underbrace{\theta^T x}_z \geq 0$

② when $\theta^T x < 0$



(18)

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

$\downarrow$ $\downarrow$ $\downarrow$
 -3 1 1

Suppose we choose parameters based on method

• $y=1$ if $\underbrace{-3 + x_1 + x_2}_{\theta^T x} \geq 0$

$$\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

Next
mein
choose
seekhna

$$x_1 + x_2 \geq +3$$

$$x_1 + x_2 = 3$$

So Regions divide ho gae! $\neq 0$

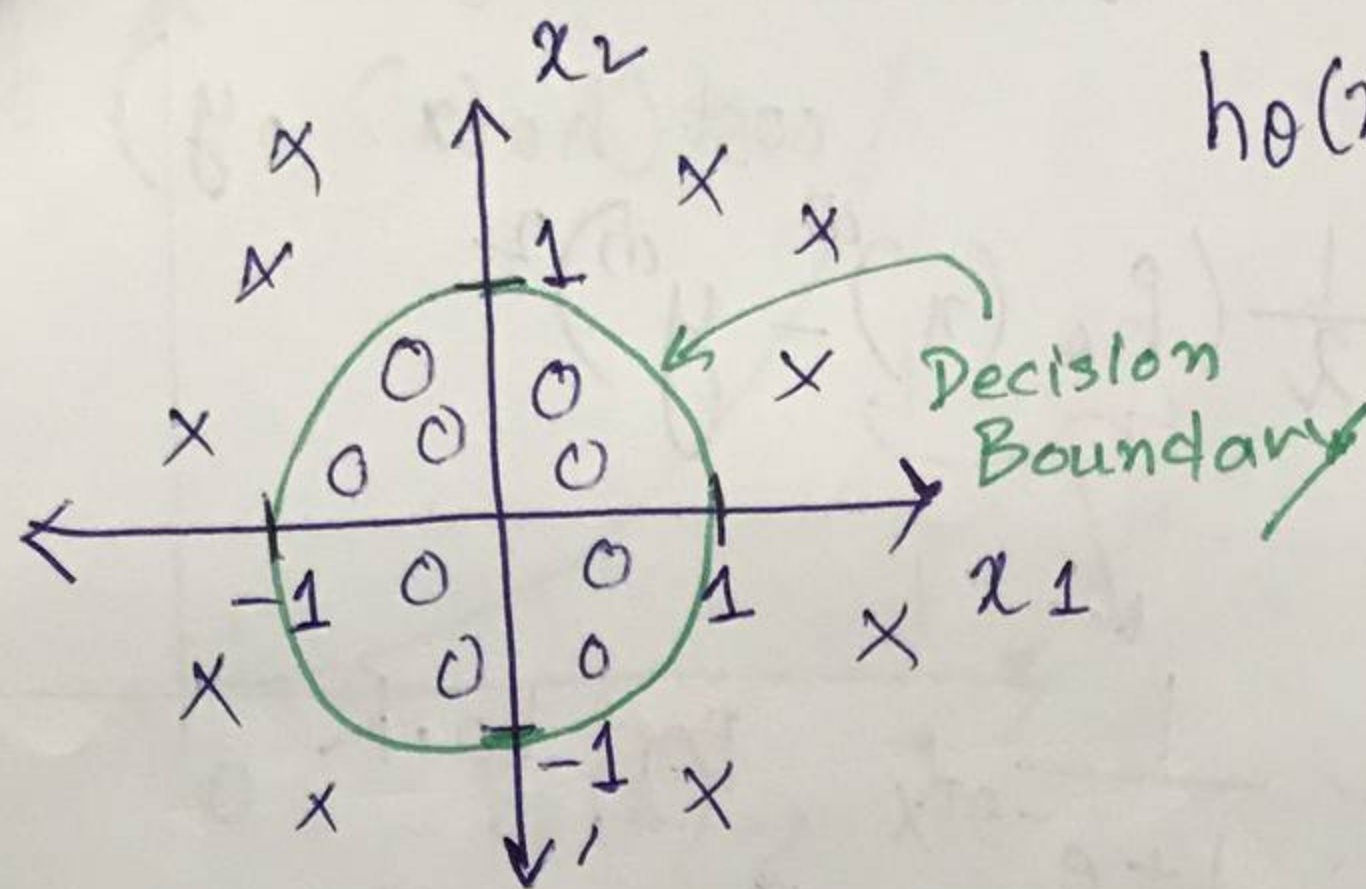
• Decision Boundary is property of hypothesis & Parameters of hypothesis

$$x_1 + x_2 = 3 \rightarrow h_{\theta}(x) = 0.5$$

$$x_1 + x_2 > 3 \rightarrow \text{hypothesis} = 1 \quad y = 1$$

$$x_1 + x_2 < 3 \rightarrow \text{hypothesis} = 0 \quad y = 0$$

Linear Decision Boundaries: 19



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2)$$

1 1

→ By some process chosen

$$y = 1 \quad \text{if} \quad -1 + x_1^2 + x_2^2 \geq 0$$

$$\theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$x_1^2 + x_2^2 \geq 1$$

eq of circle —

6.4 Fitting Parameters:

COST FUNCTION:

Training Ex $\rightarrow m$

Features $\rightarrow n$

$$x_0 = 1$$

$$y \in \{0, 1\}$$

$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose parameters θ ?

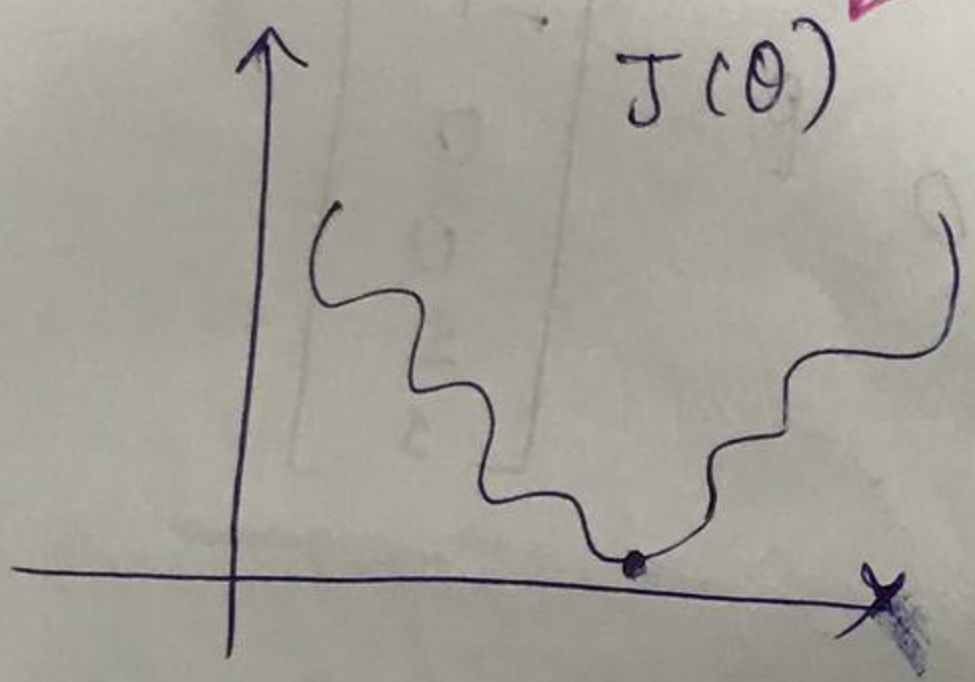


(20)

L.R. $\rightarrow$ $J(\theta) = \frac{1}{n} \sum \frac{1}{2} [h_{\theta}(x)^{(i)} - y^{(i)}]^2$
Lo.R. $\rightarrow$

$\text{cost}(h_{\theta}(x)^{(i)}, y^{(i)}) = \frac{1}{2} (h_{\theta}(x)^{(i)} - y^{(i)})^2$
 $\text{cost}(h_{\theta}(x), y)$

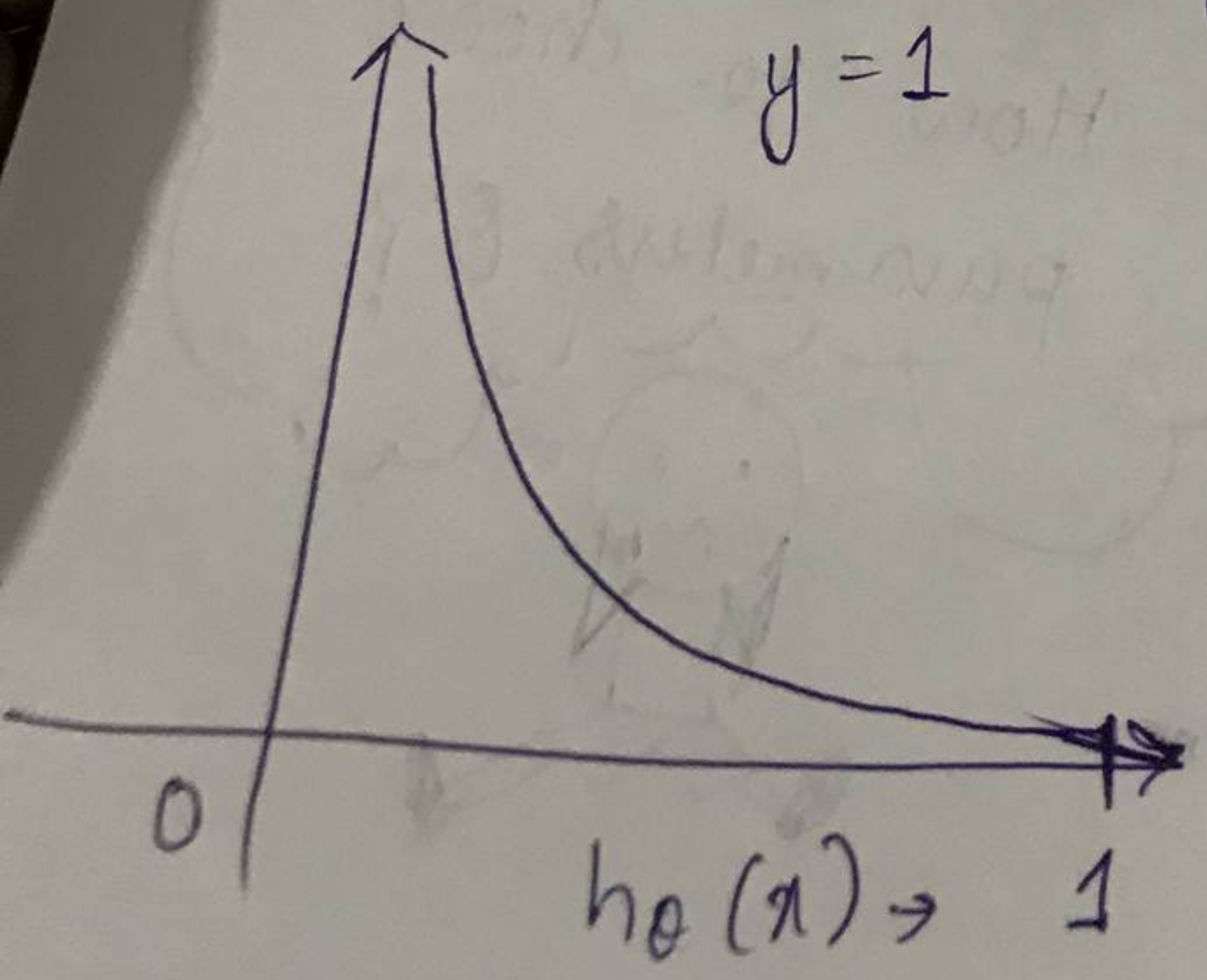
If use G.D on it
 $\rightarrow$ Non-Convex



$\frac{1}{1 + e^{-\theta^T x}}$ For Lo.R
Convex



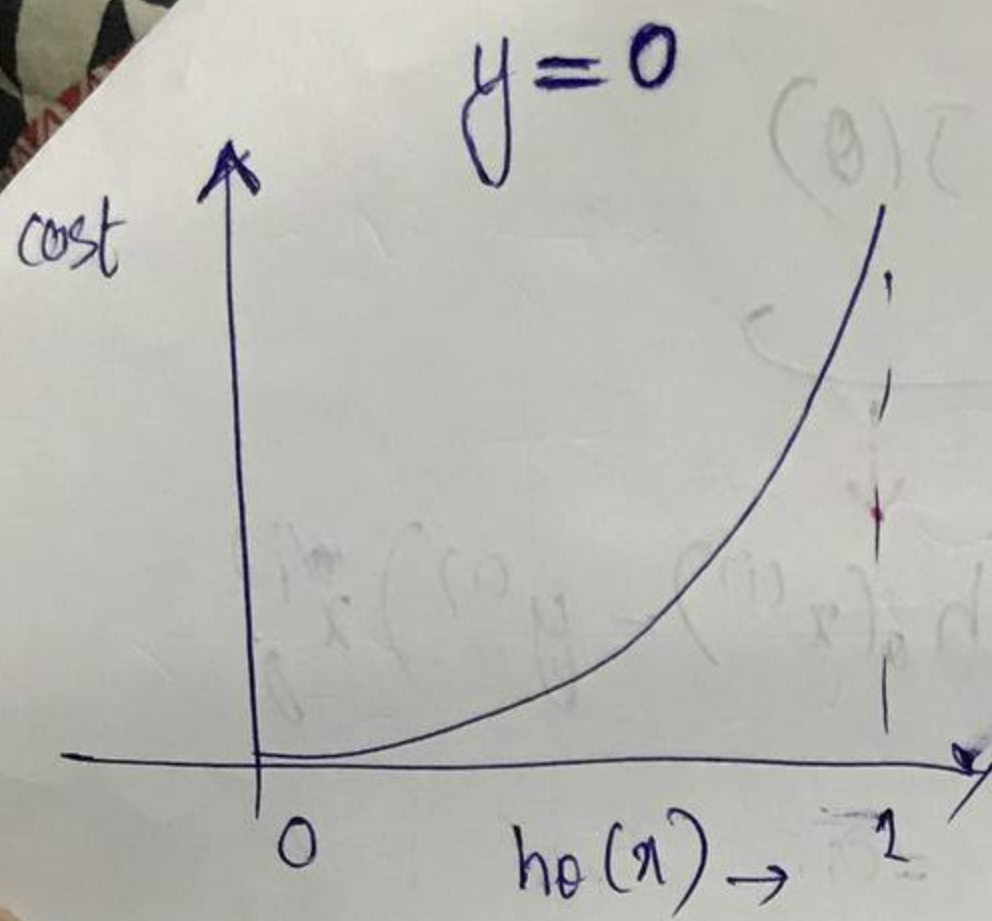
$\text{cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & y=1 \\ -\log(1 - h_{\theta}(x)) & y=0 \end{cases}$



cost = 0 if $y=1, h_{\theta}(x)=1$

$h_{\theta}(x) \rightarrow 0$
cost $\rightarrow \infty$

(21)



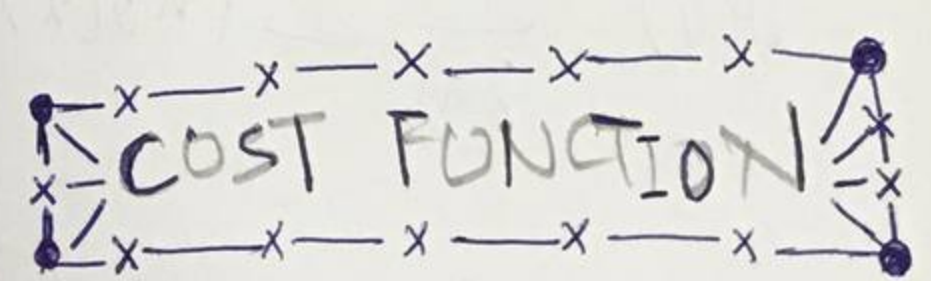
1.5

Simplified Cost Func + G.D

$y=0, 1$ always

$$\text{cost}(h_0(x), y) = -y \log(h_0(x)) - (1-y) \log(1-h_0(x))$$

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_0(x^{(i)}) + (1-y^{(i)}) \log(1-h_0(x^{(i)})) \right]$$



1.2

To fit parameters θ :

$$\min_{\theta} J(\theta)$$

To make a prediction given a new x :

$$h_0(x) = \frac{1}{1 + e^{-\theta^T x}}$$

(22)

Repeat $\left\{ \theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta) \right.$ $0 = 0$

$\sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)}$

Same as Linear Regression $= 0$

Diff is $\rightarrow h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$

6.6

Tired Baaki Next Time :C

Regularization

data $\xrightarrow{\text{train}}$ accuracy very high
test data $\rightarrow$ acc. not too high

Data is overfitting

$\rightarrow$ So we regularize data to avoid this

$$J(\theta) = \frac{1}{2m} \sum (h_{\theta}(x)^i - y^i)^2 + \lambda \sum_{j=1}^m \theta_j^2$$

$\lambda \rightarrow$ very small overfit
 $\lambda \rightarrow 0$ large underfit

add this factor